Low-Code Quantum Application Development: From Visual Modeling to Real Quantum Execution

IAAS Tutorial

University of Stuttgart

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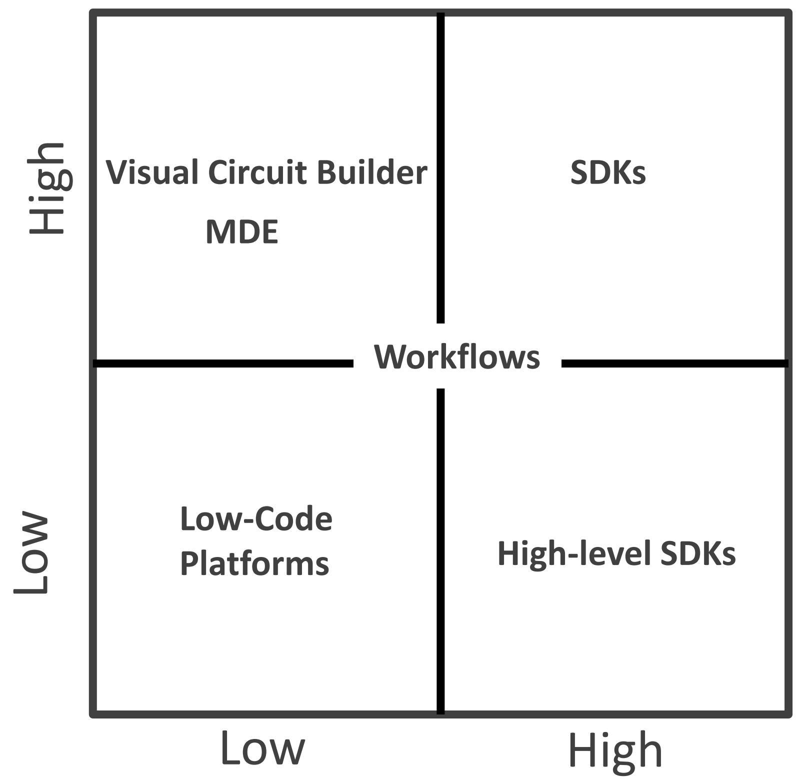
Institute for Architecture of Application Systems

Motivation

Quantum Computing Knowledge



Programming Knowledge

Motivation

- Existing quantum low-code platforms:
  - Often closed-source
  - Vendor-locked
  - Circuit-level only

Research Question

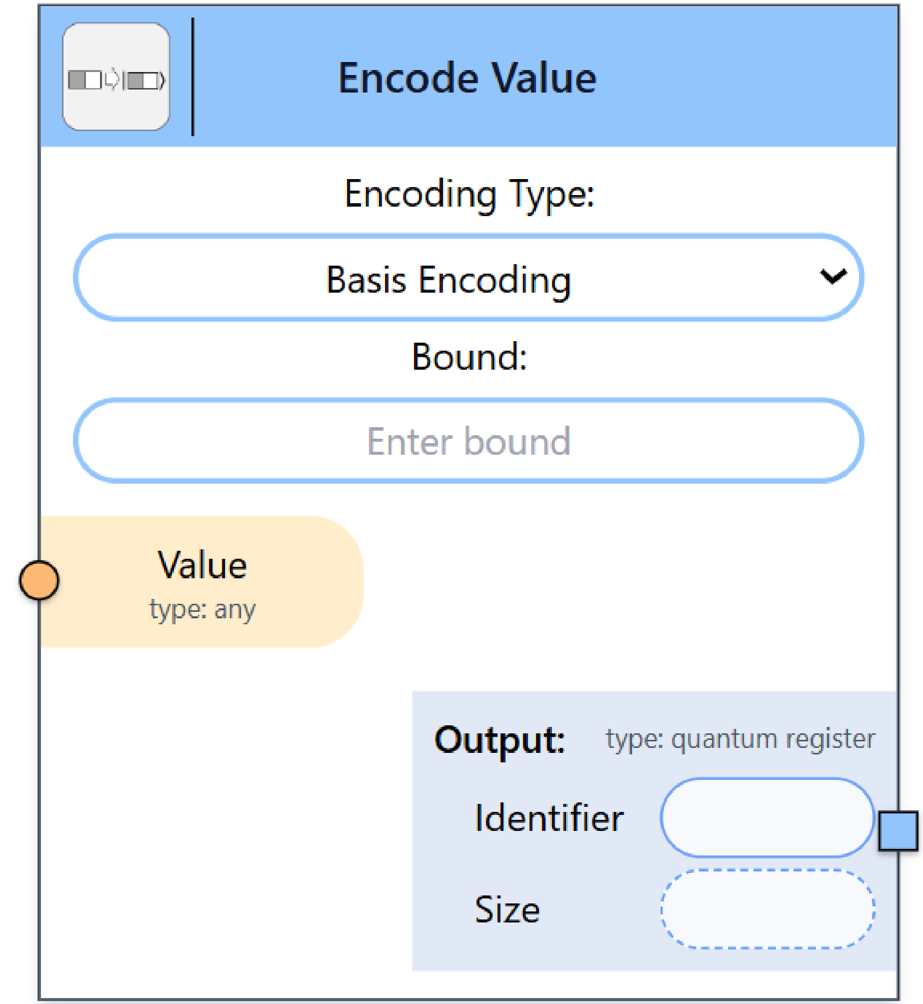
- To make quantum computing accessible:
  - Users need visual abstractions instead of low-level coding
  - Hybrid quantum-classical operations must be represented intuitively
  - Different expertise levels must be supported
  - Models should remain reusable and extensible

<u>Research Question:</u>

Which **abstractions** and **modeling elements** are required to support modeling of quantum algorithms in a **low-code platform**?

Modeling Language

- Icon
  - Basis Encoding Pattern Icon
- Type
  - State Preparation
- Properties
  - Encoding Type
  - Bound
  - Implementation (optional)
- Inputs
  - Value
- Outputs
  - Quantum Register



Modeling Language

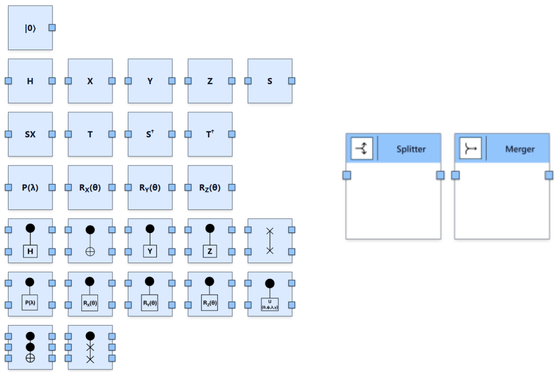
Modeling Elements

- Circuit Primitives
- Data Types
- Boundary Elements
- Operators
- Control Flow Elements
- Custom Elements

Tutorial

Modeling Language

Circuit Primitives



Data Types

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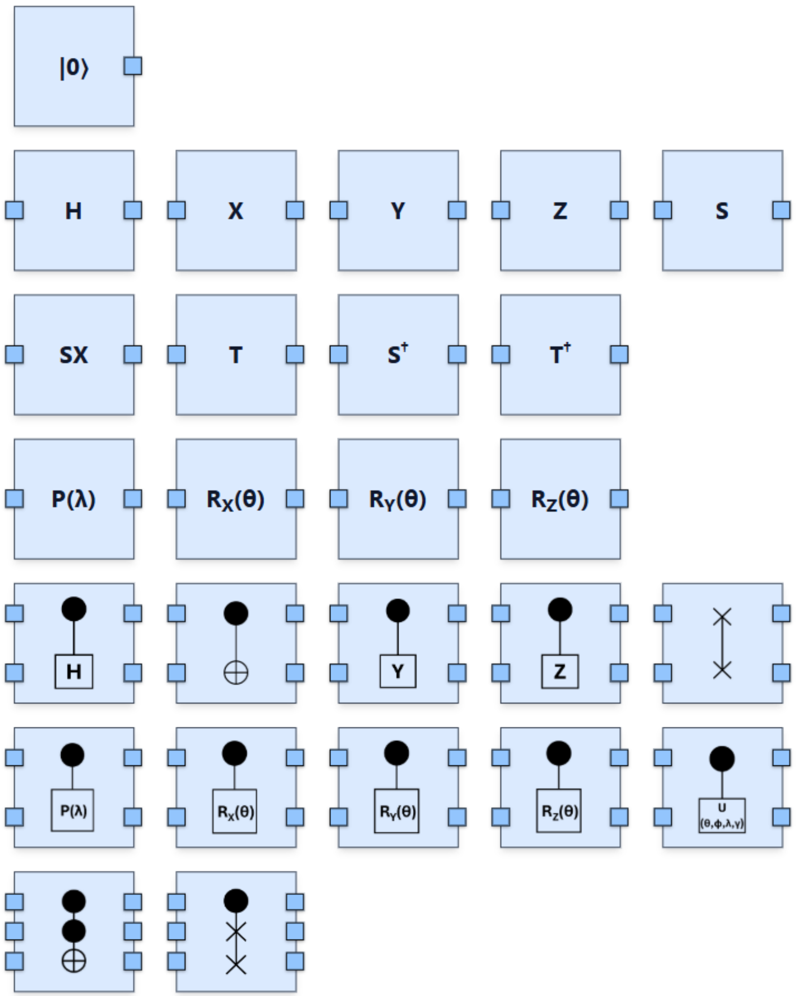

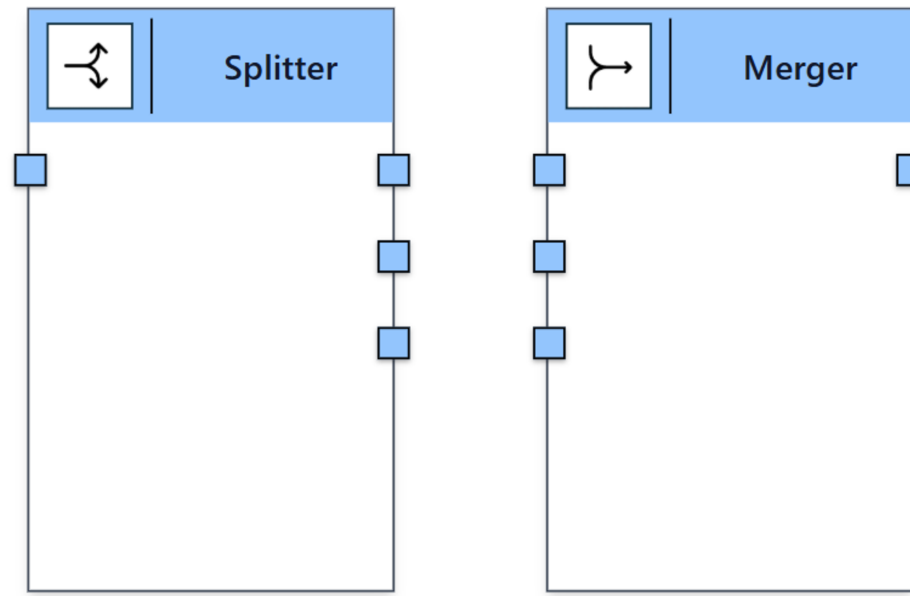

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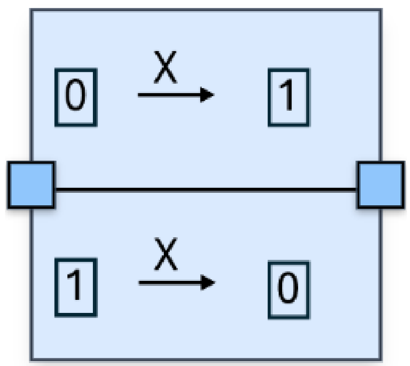
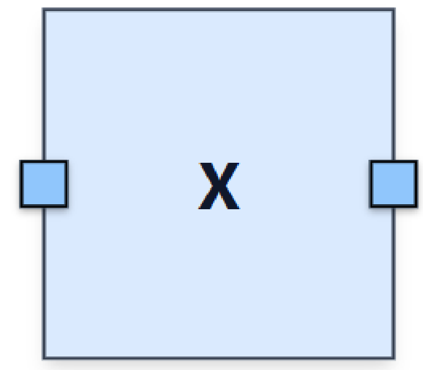


Explorer



Pioneer

Modeling Language

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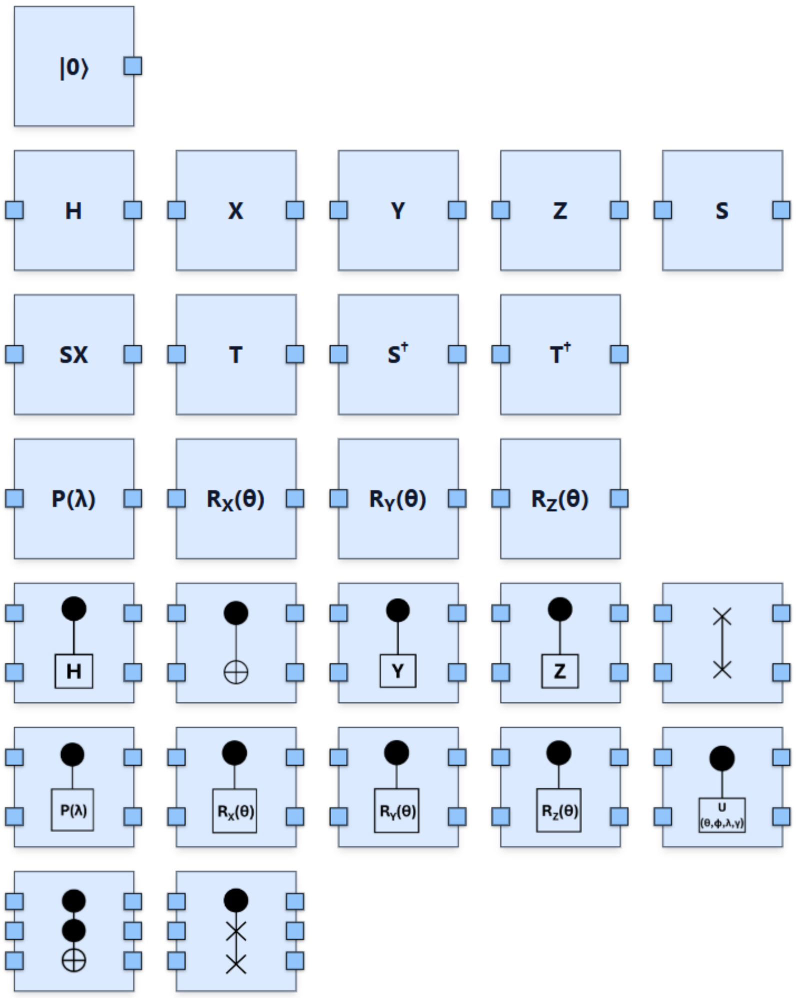

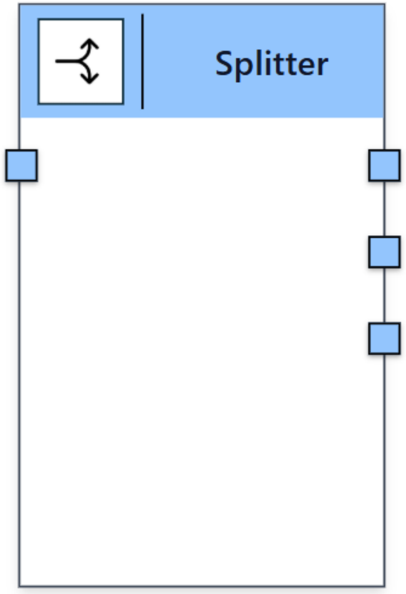

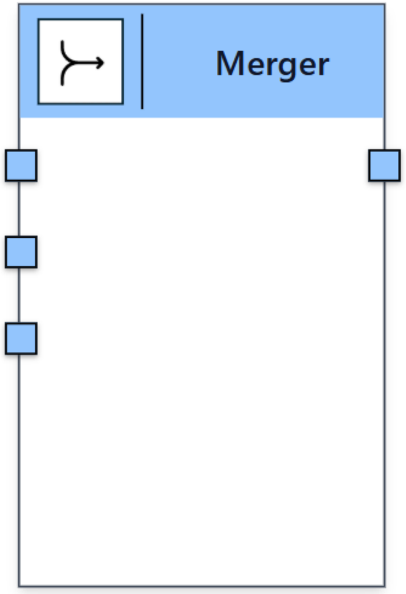

Modeling Language

Tutorial

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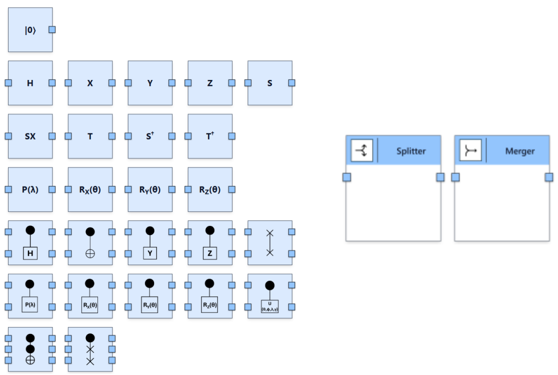

Data Types

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Custom Elements

Modeling Language

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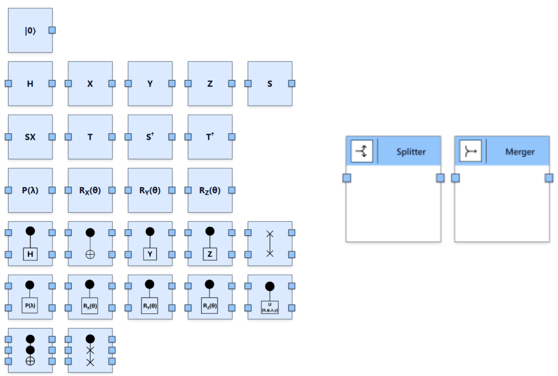

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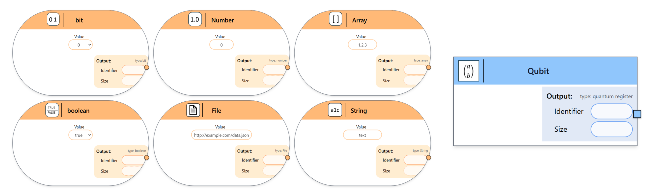

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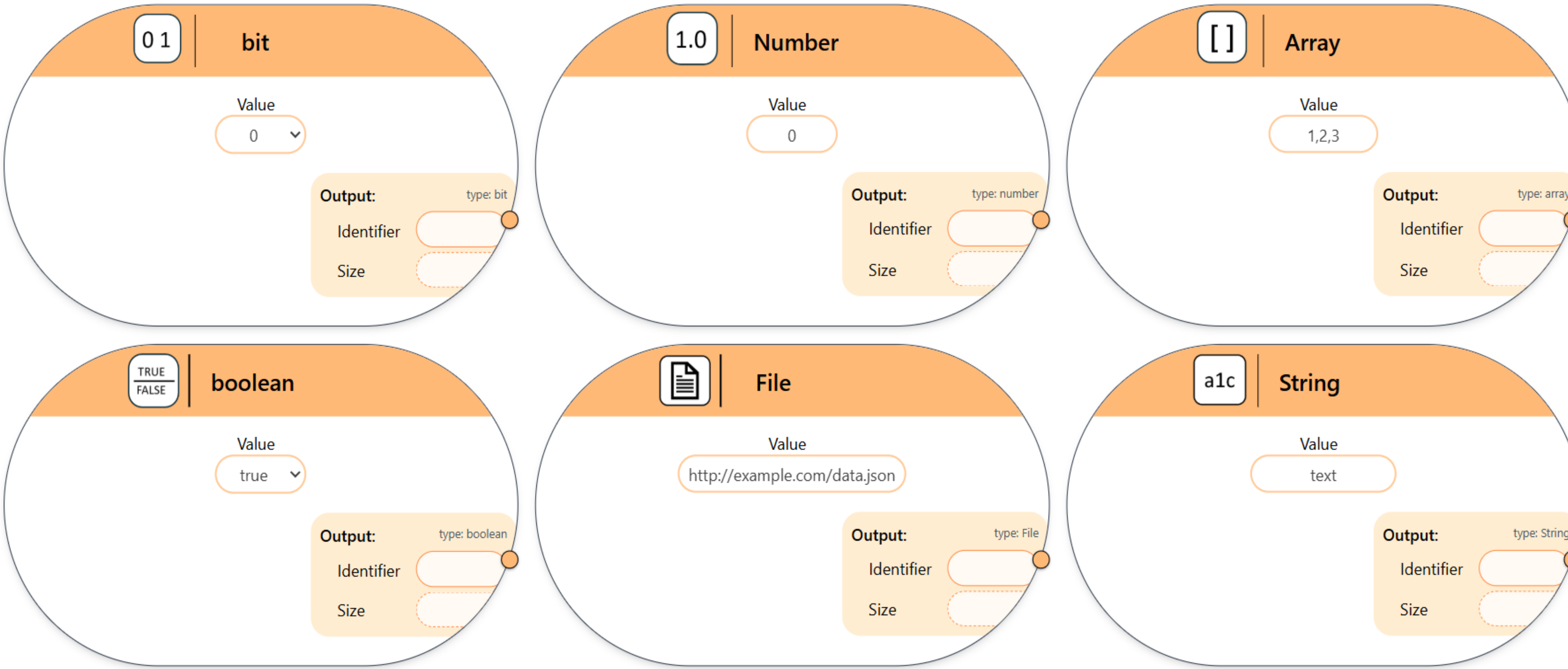

Modeling Language

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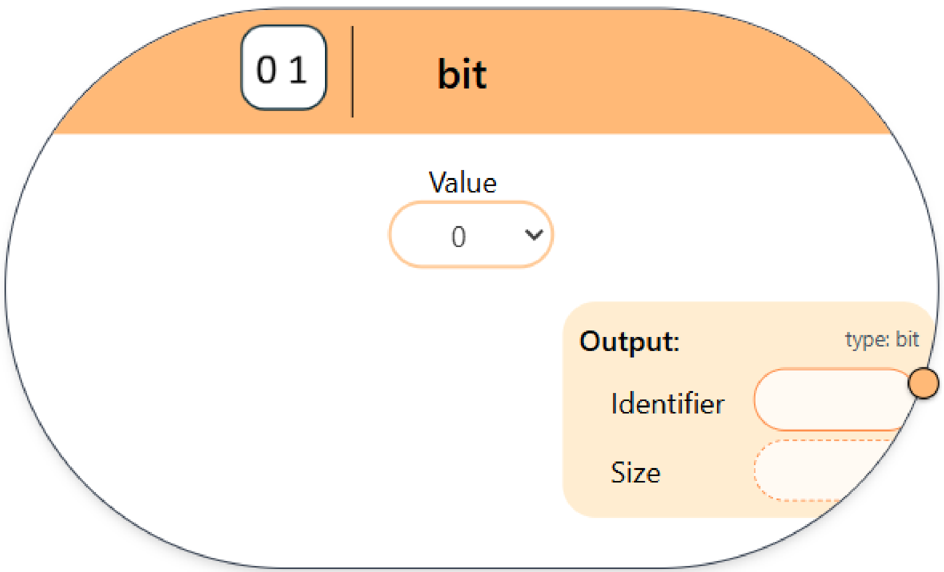

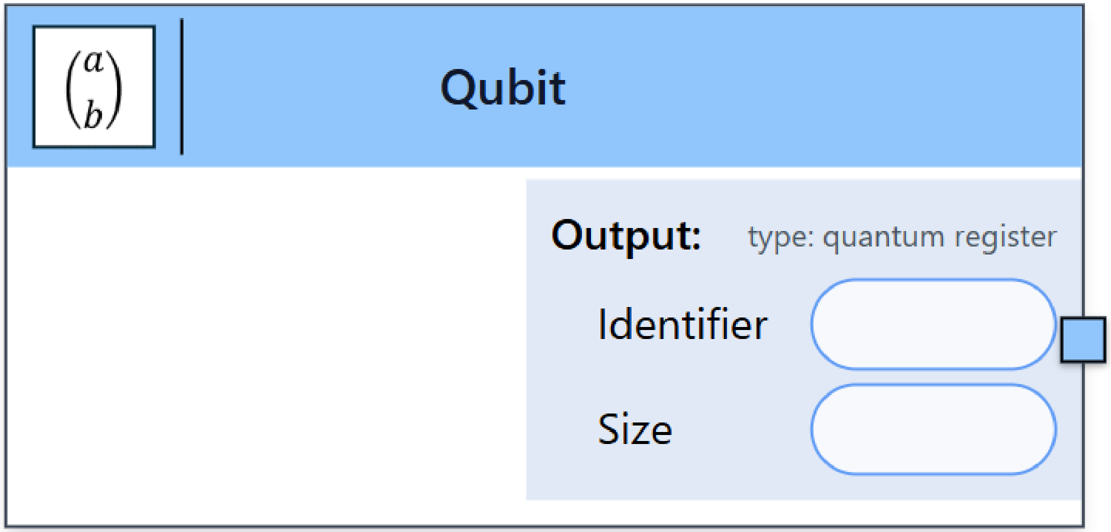

Modeling Language

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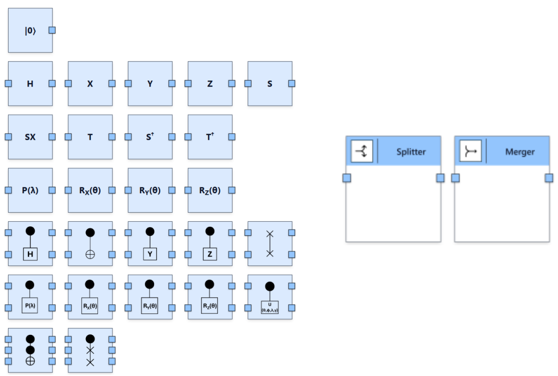

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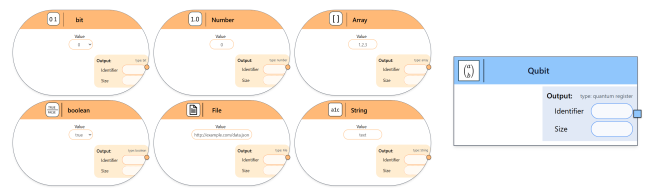

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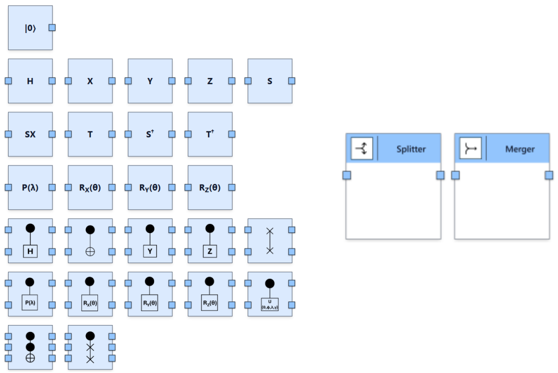

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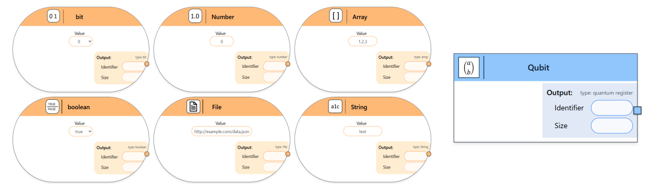

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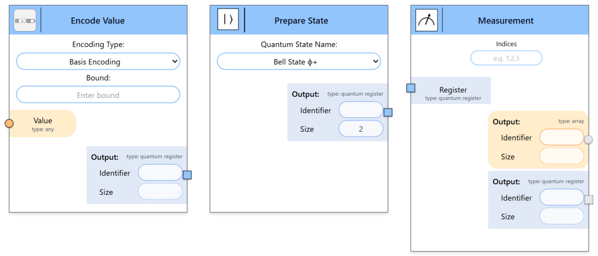

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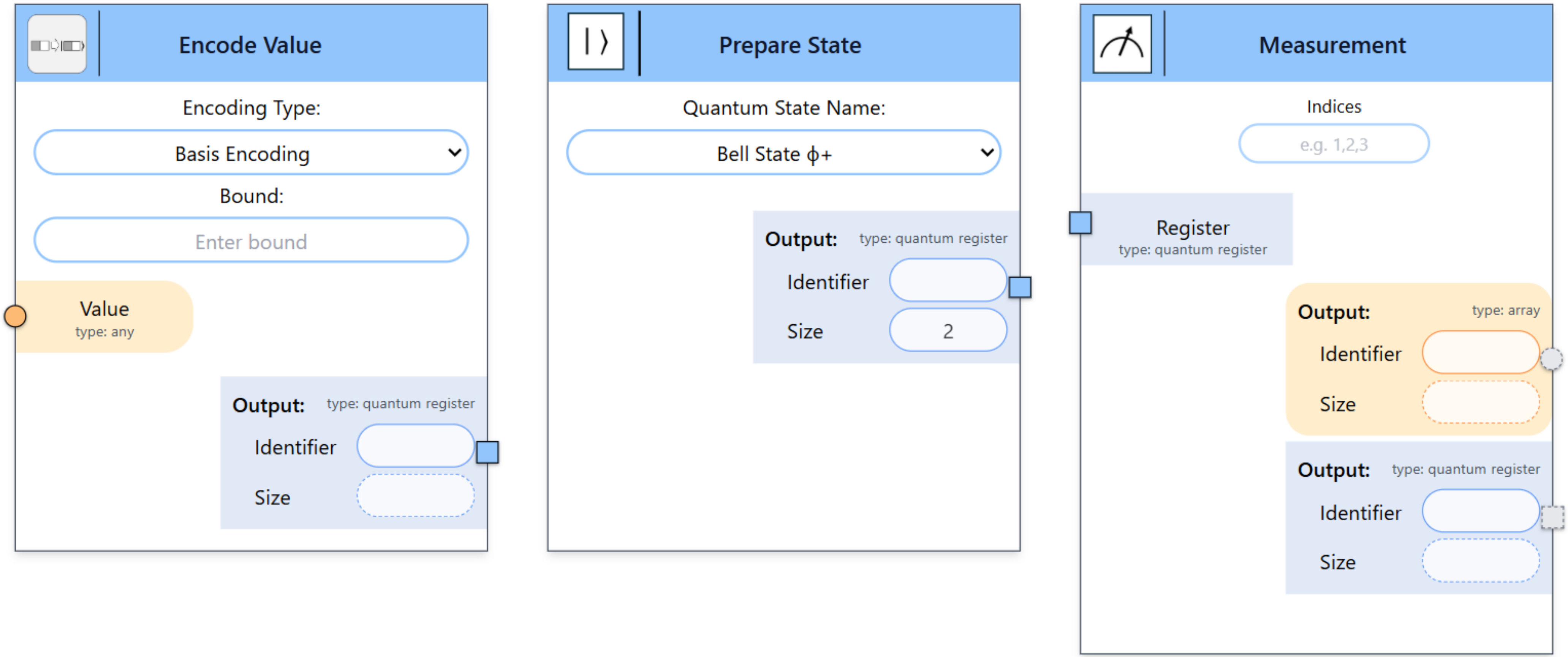

Modeling Language

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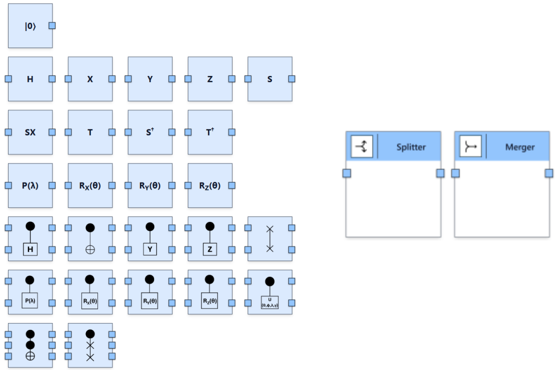

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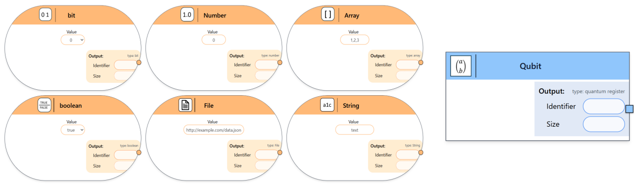

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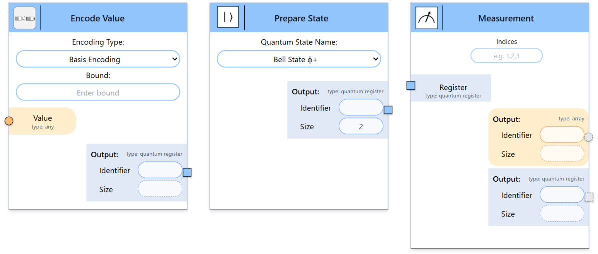

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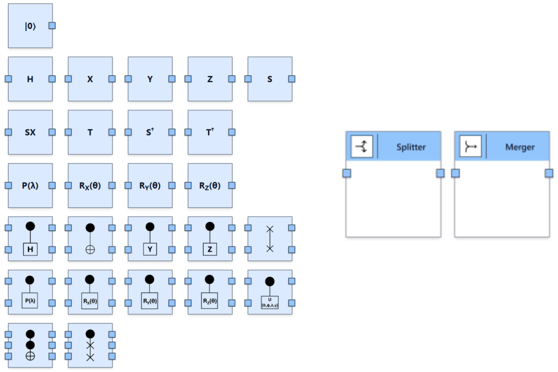

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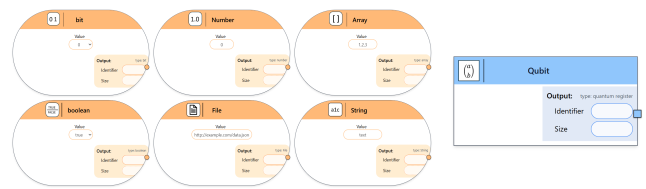

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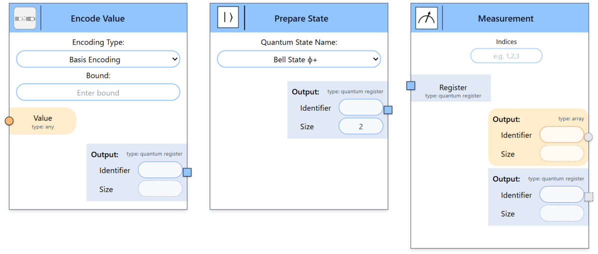

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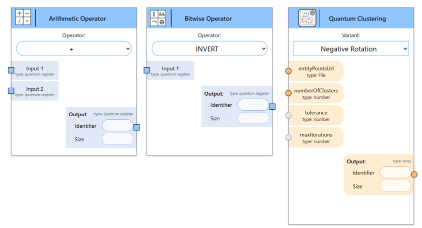


Operators

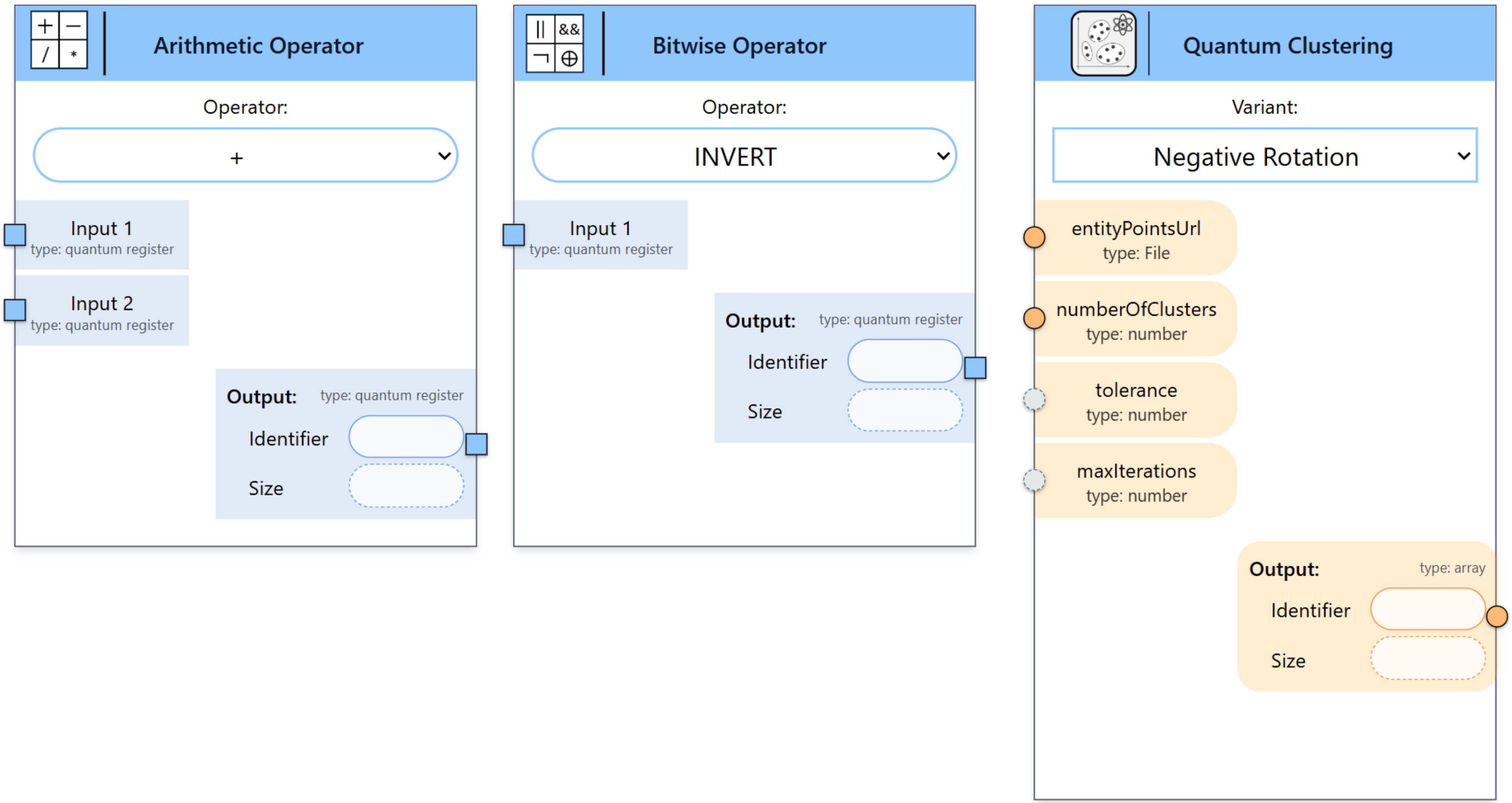
Control Flow Elements

Custom Elements

Tutorial

Modeling Language

- **Operatoren:**



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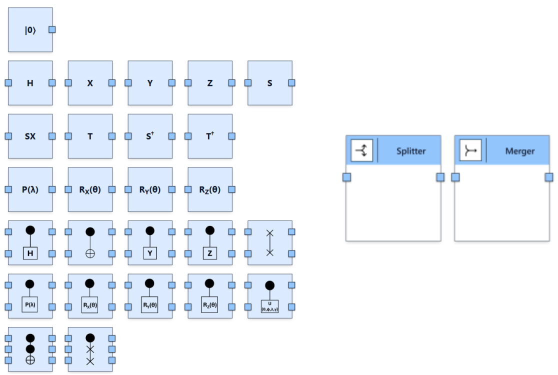
Modeling Language

Circuit Primitives



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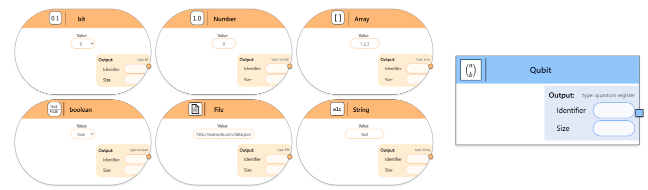

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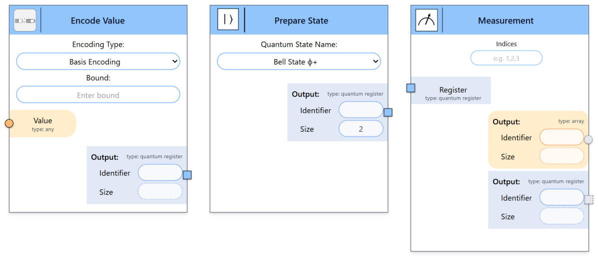

Modeling Elements

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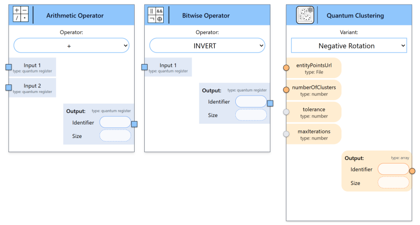

Control Flow Elements

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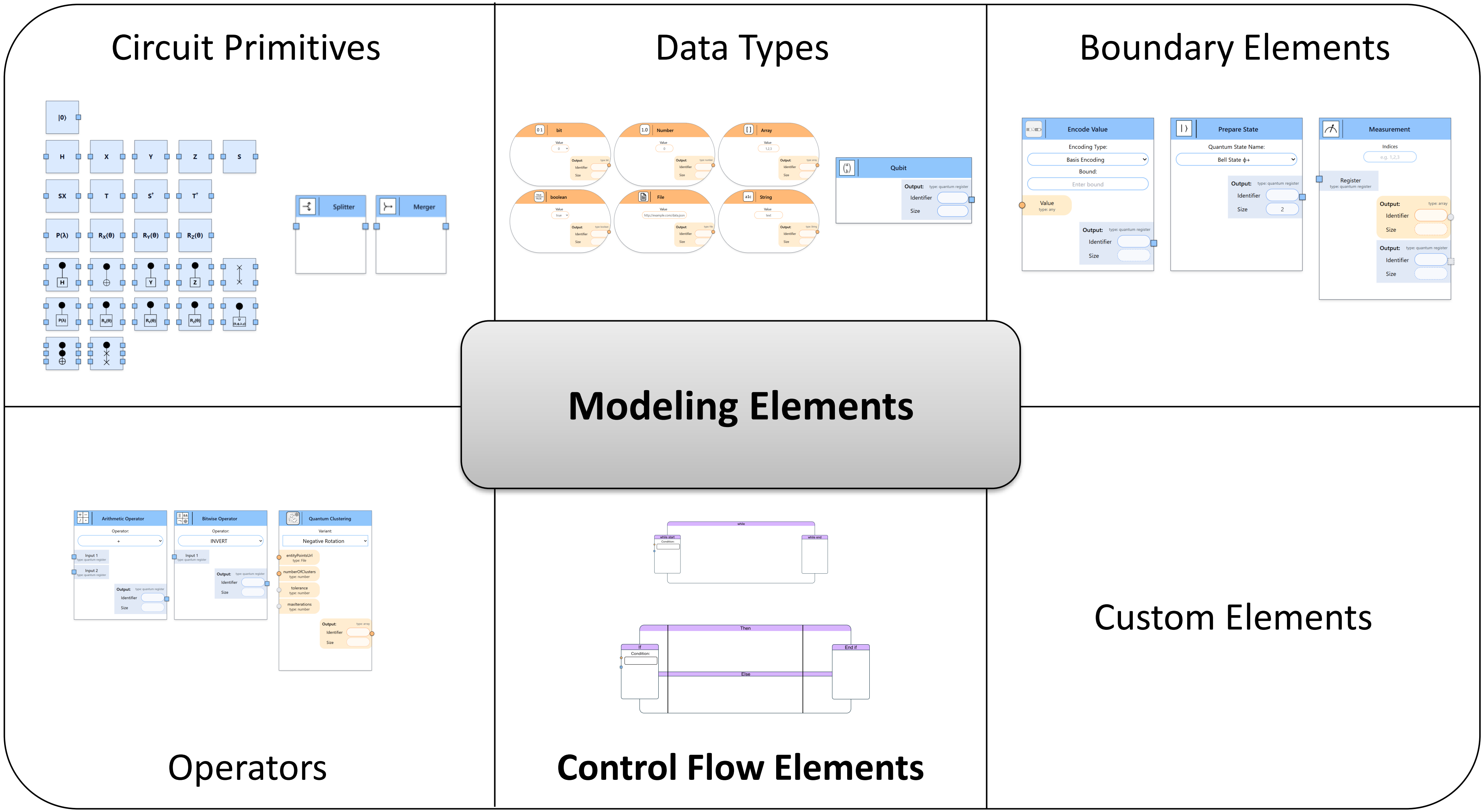

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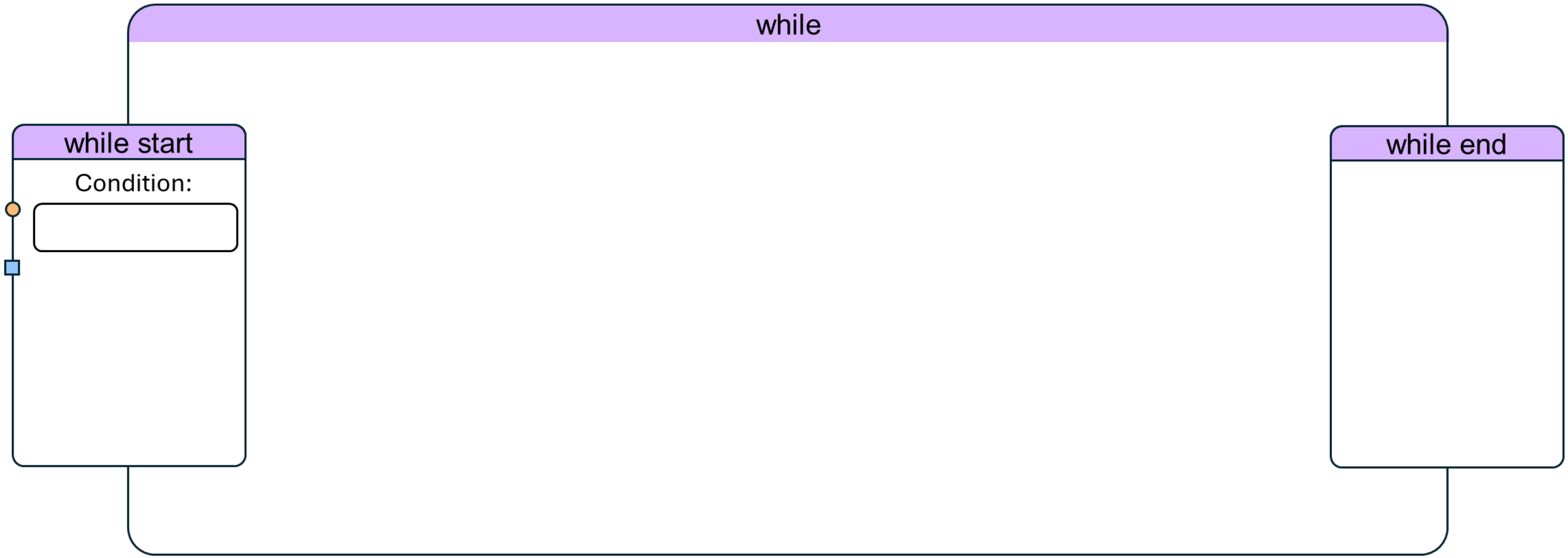

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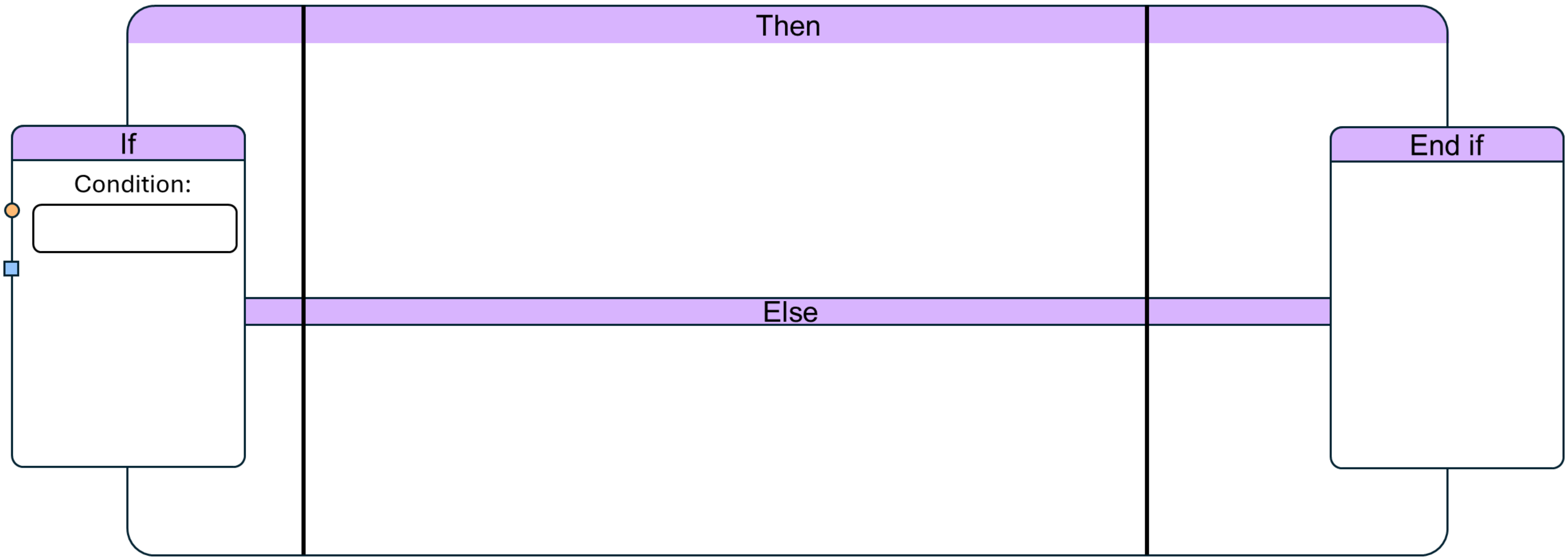

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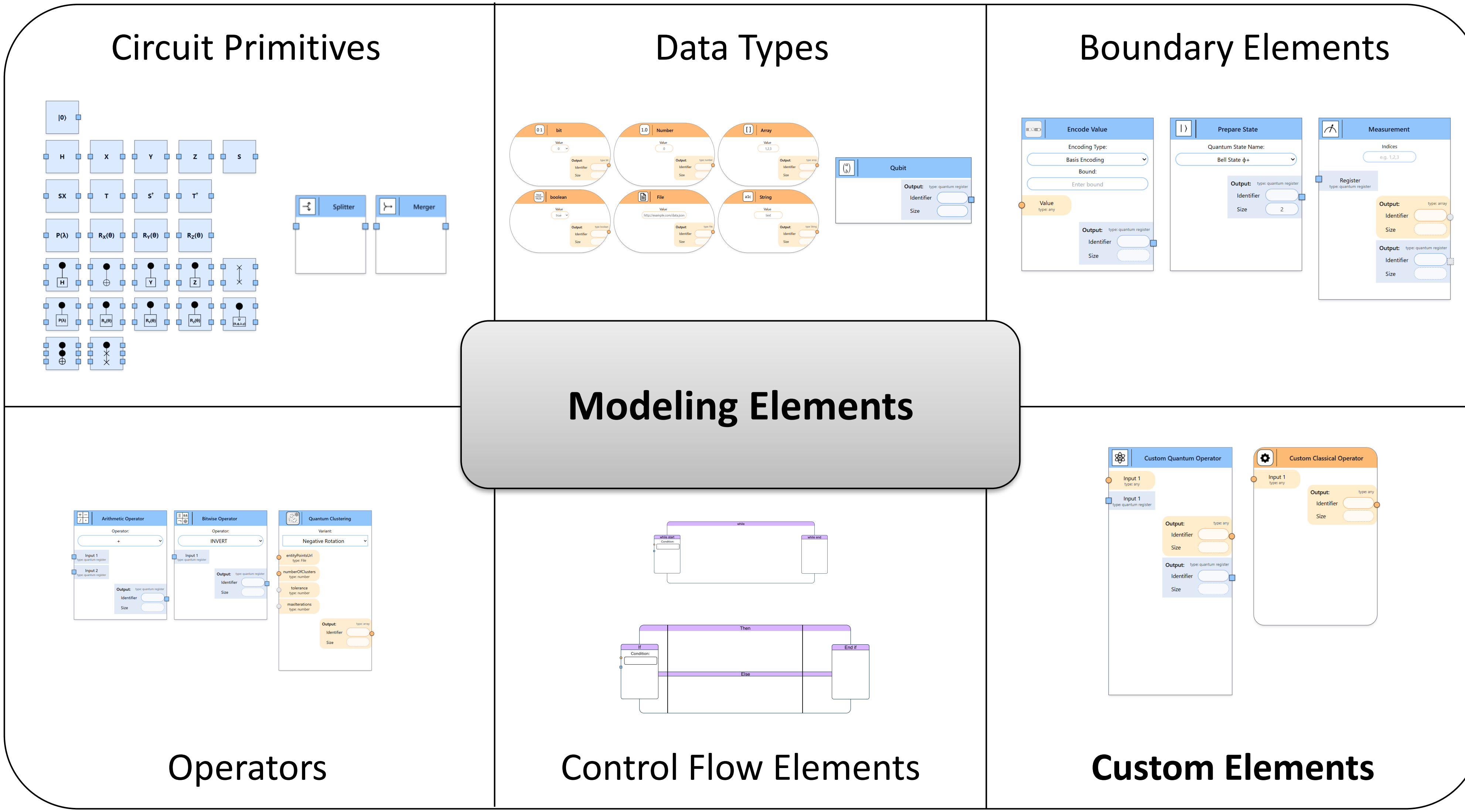
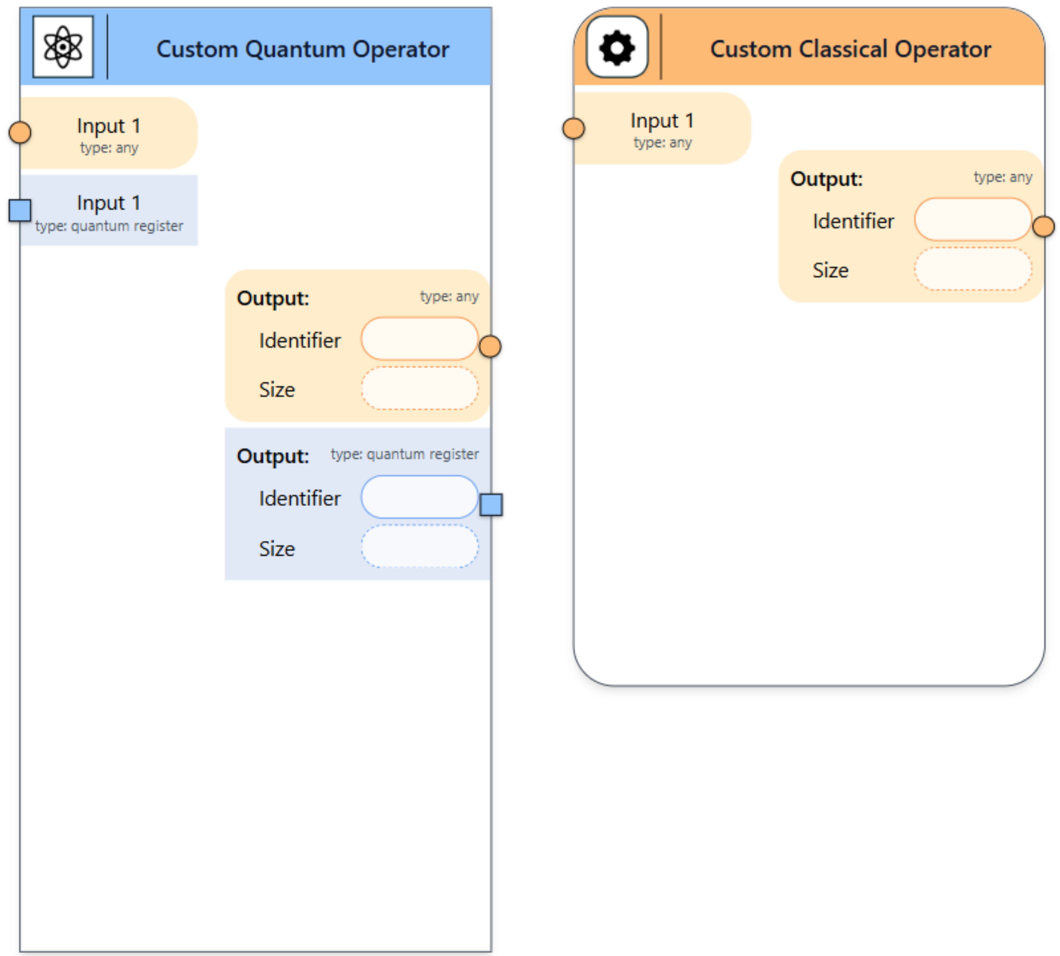


Tutorial

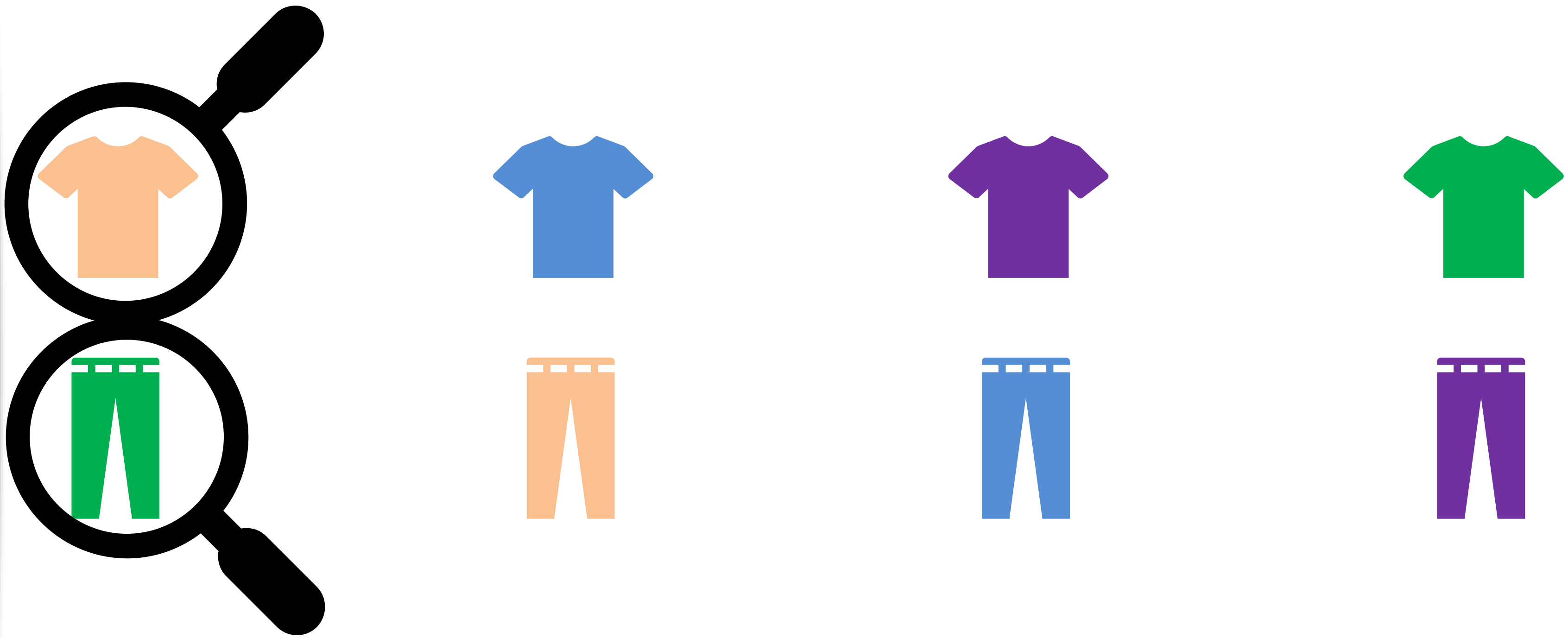
Modeling Language

- **Custom Elements:**



Task for Today

- Classical Solution:



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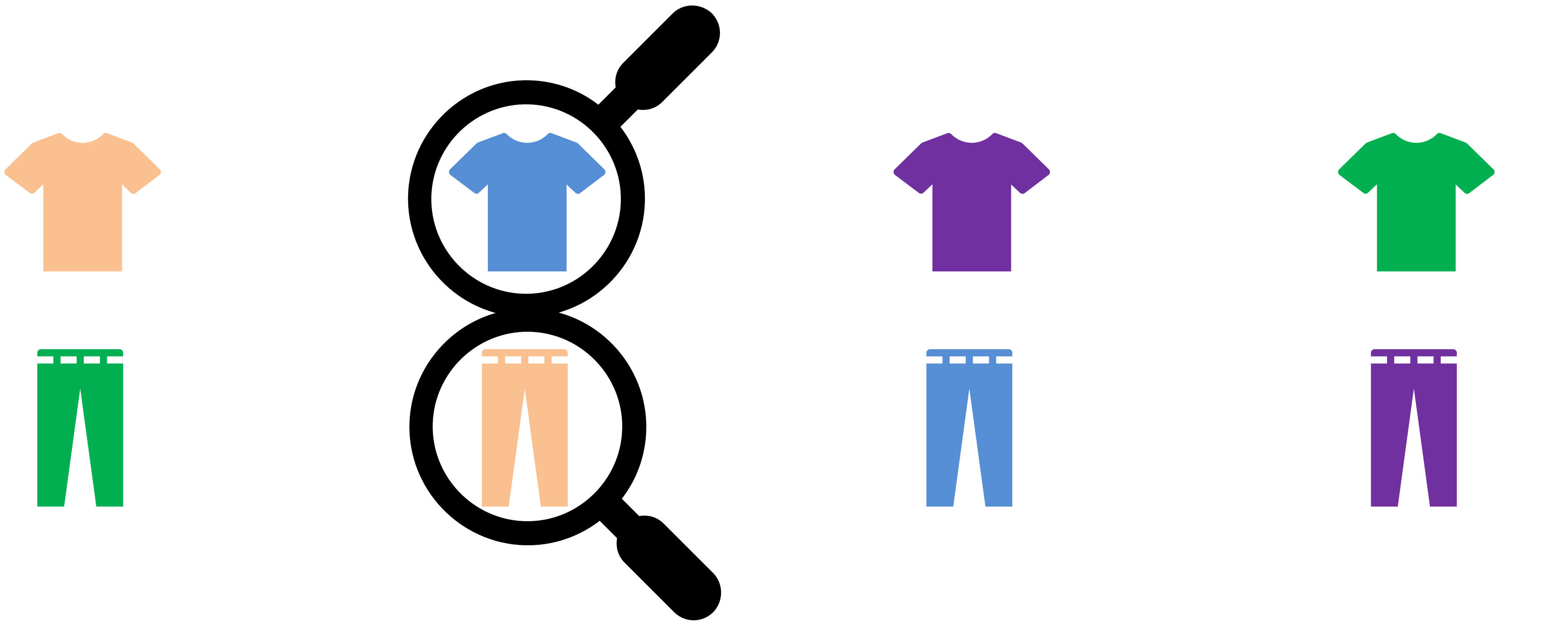

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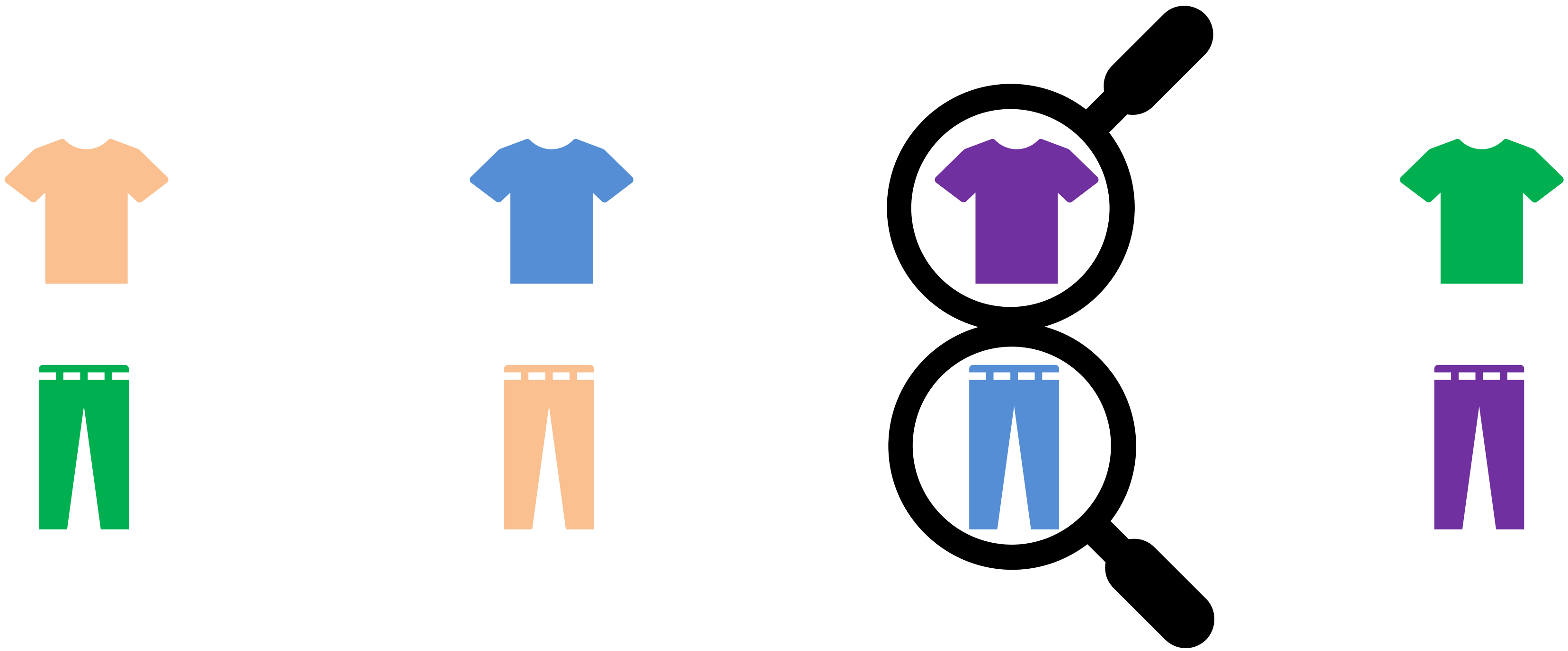

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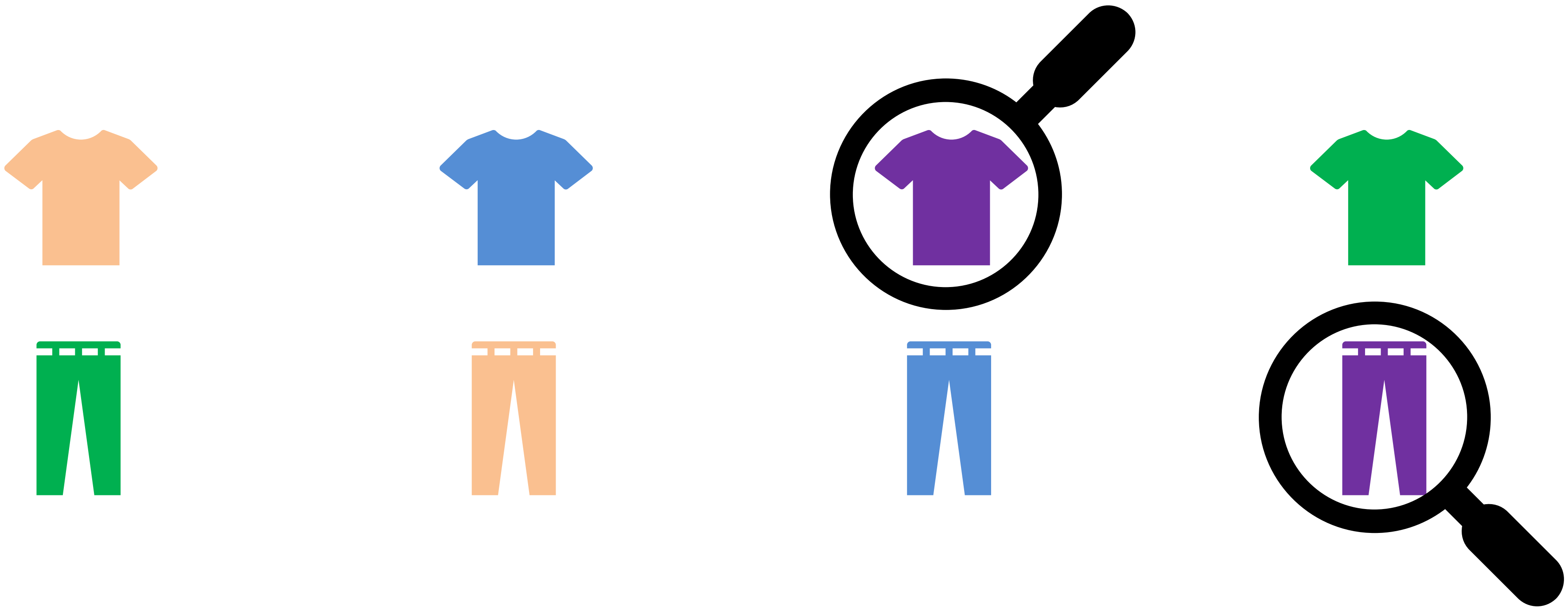

Task for Today

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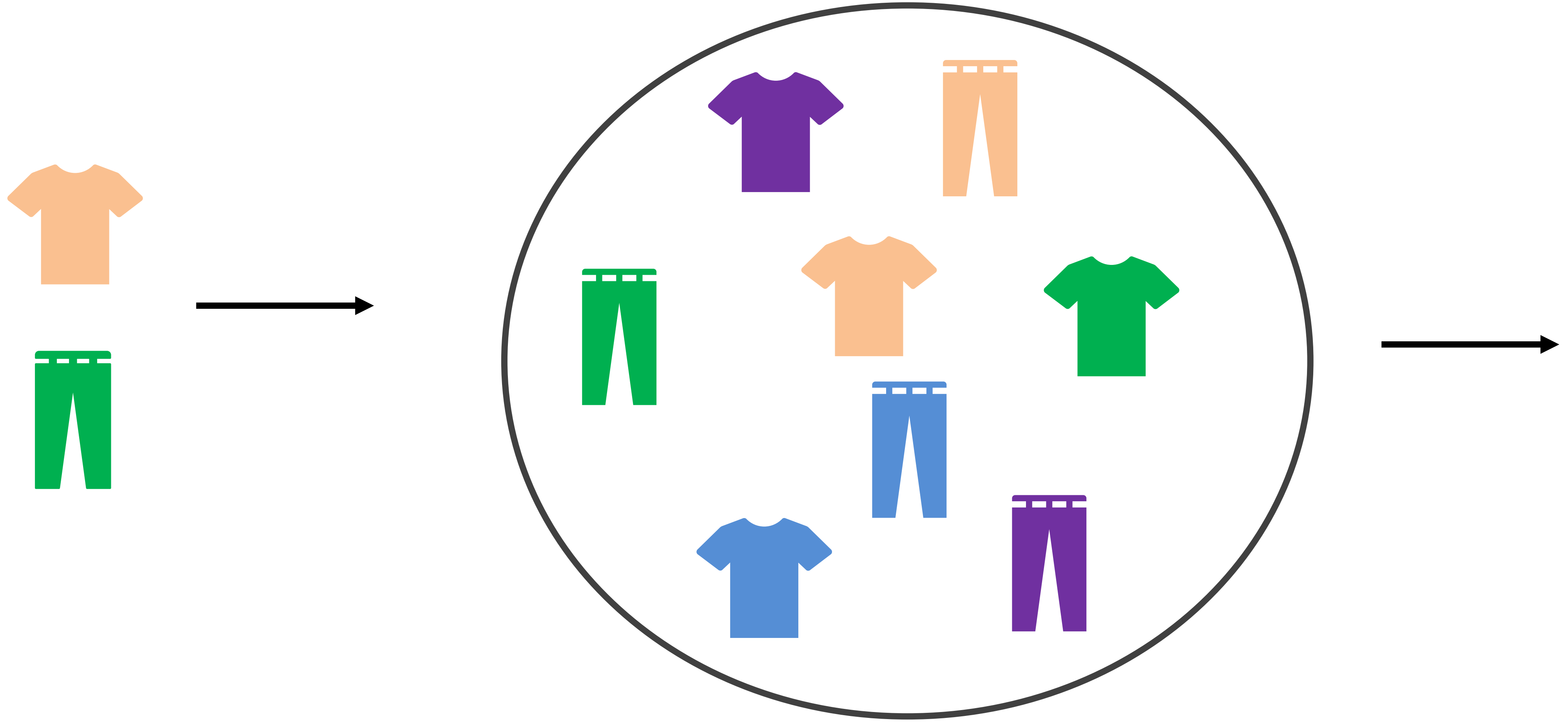

Uniform Superposition

Task for Today

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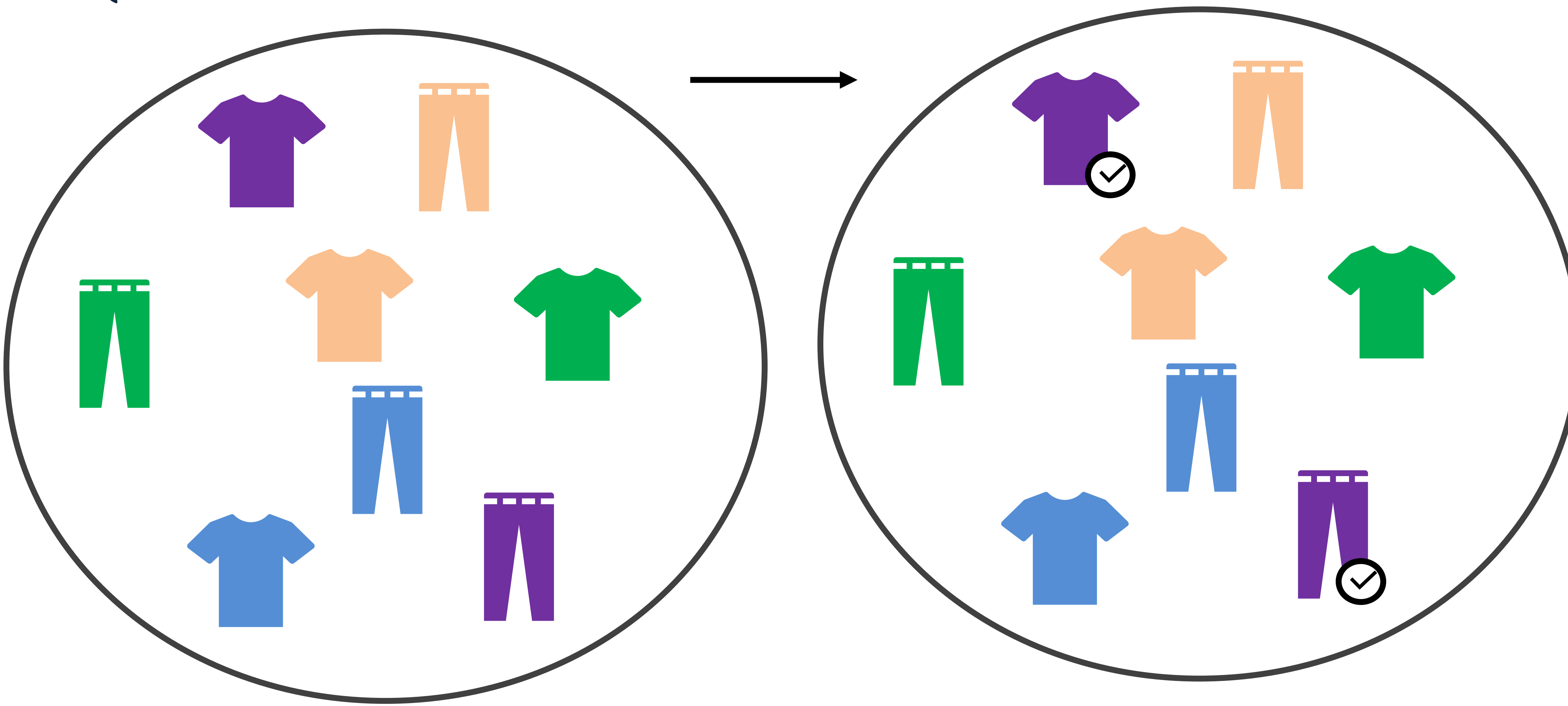

Uniform Superposition

Oracle

Task for Today

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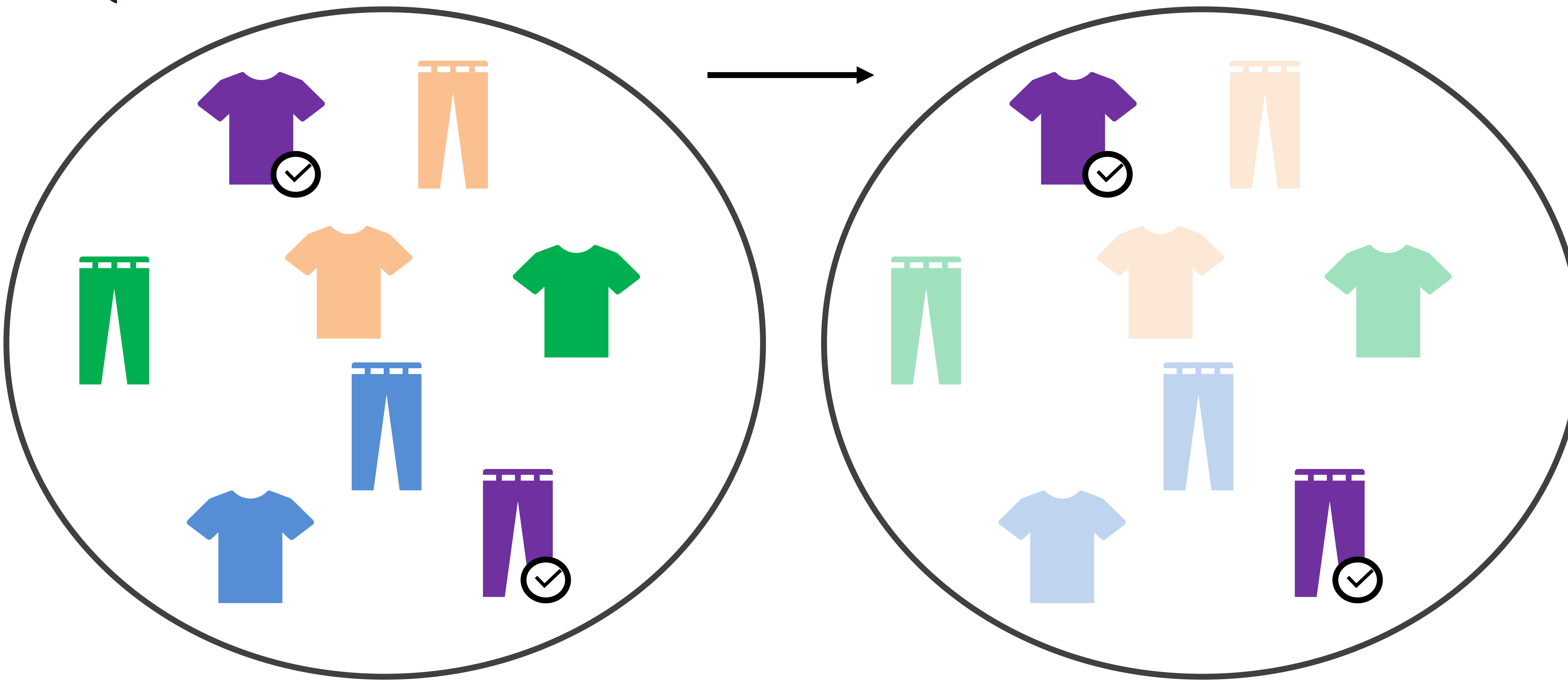
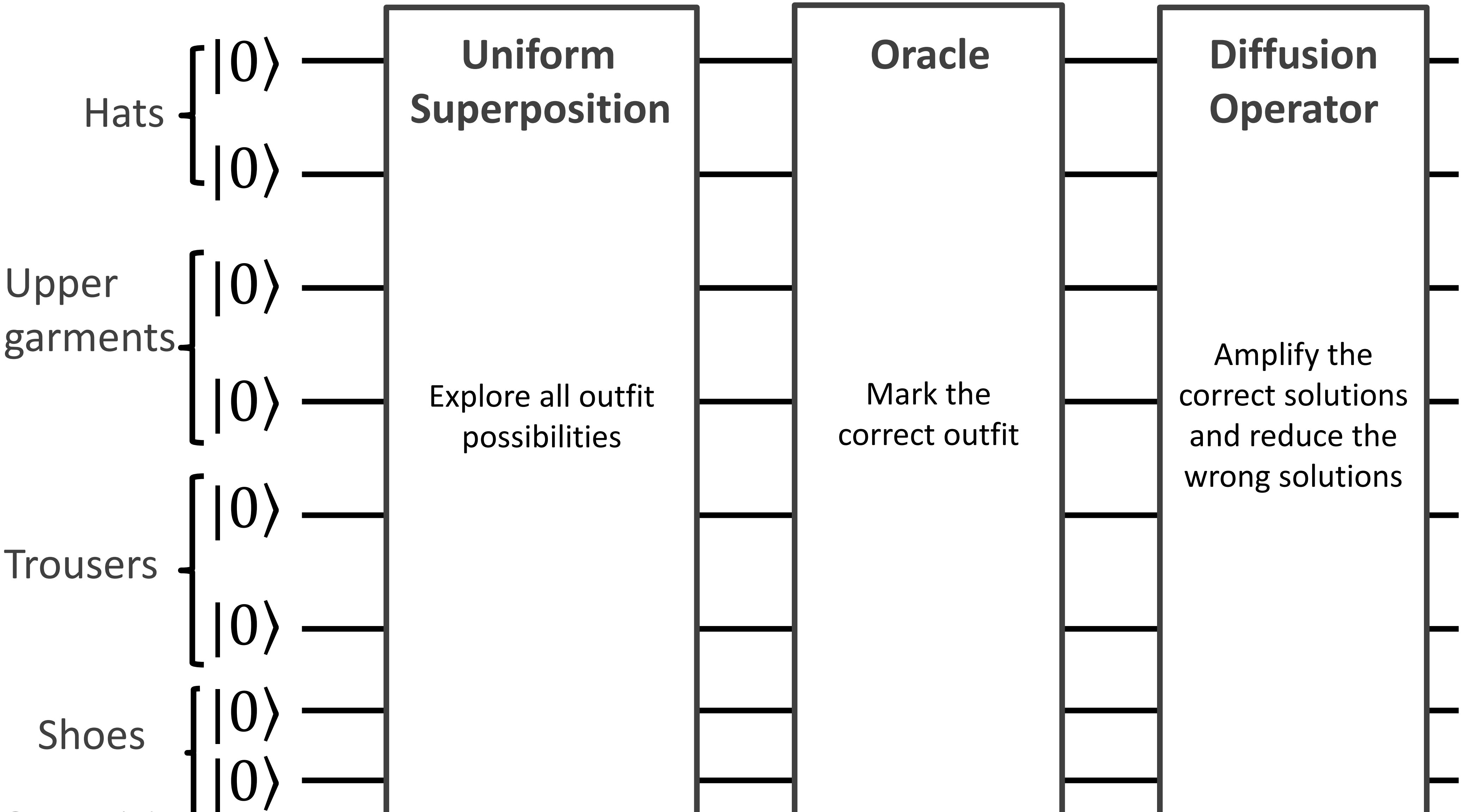


Oracle

Diffusion Operator

Task for Today - Grover‘s Search Algorithm



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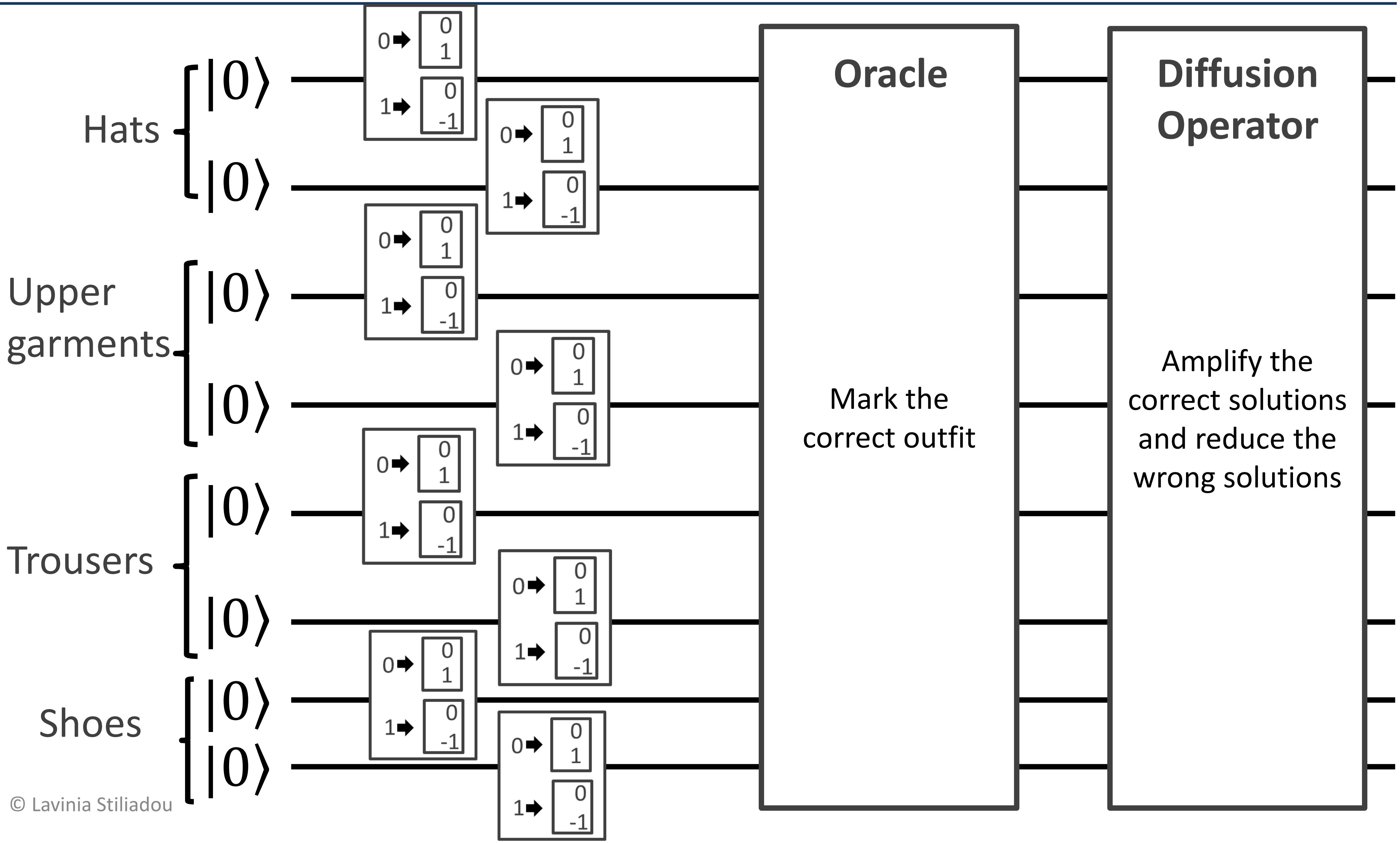

Task for Today - Grover‘s Search Algorithm Oracle

- The oracle should mark the correct solution
- We have 8 qubits

- If we want always the second option:
  - Second hat → 01
  - Second upper garment → 01
  - Second trouser →01
  - Second shoe → 01

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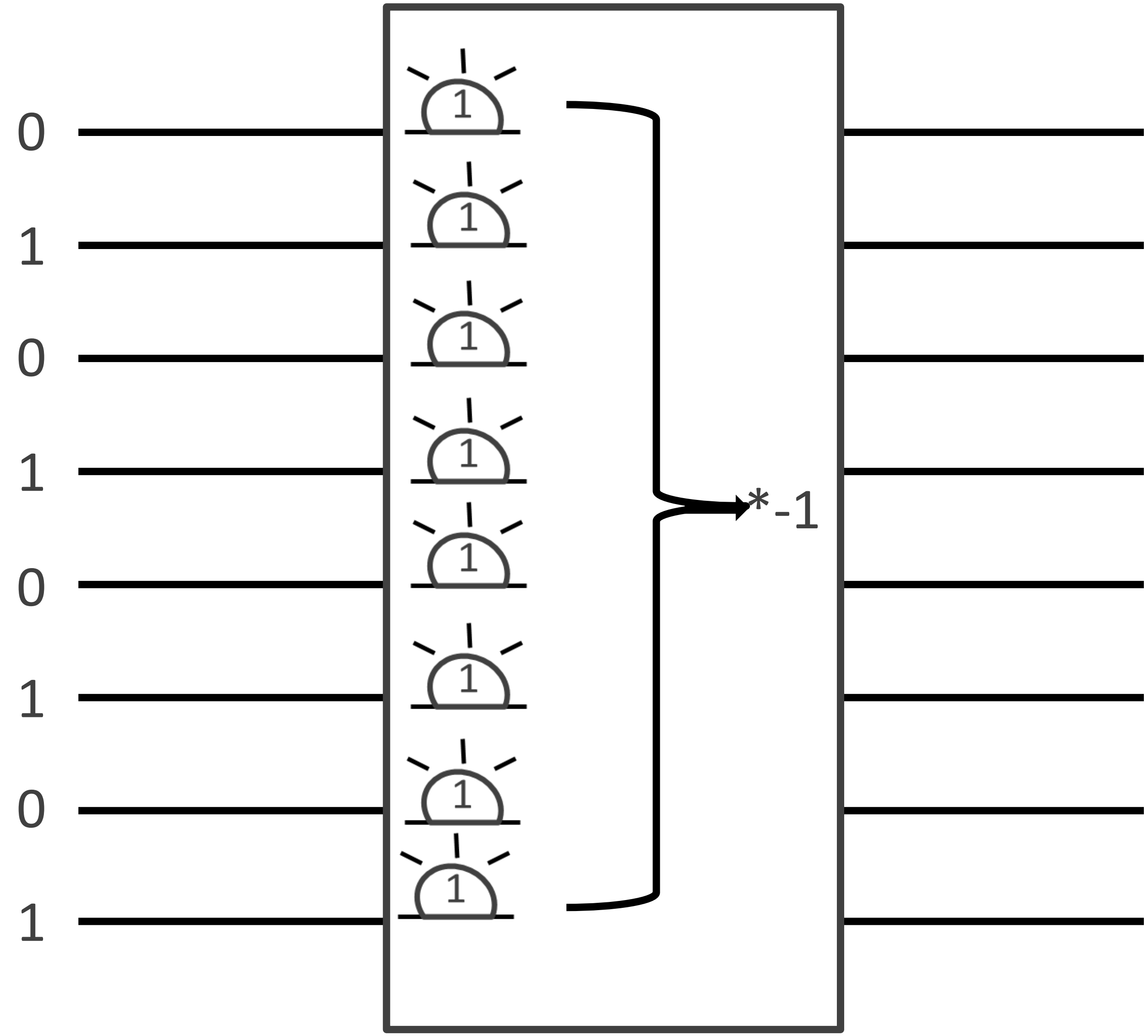

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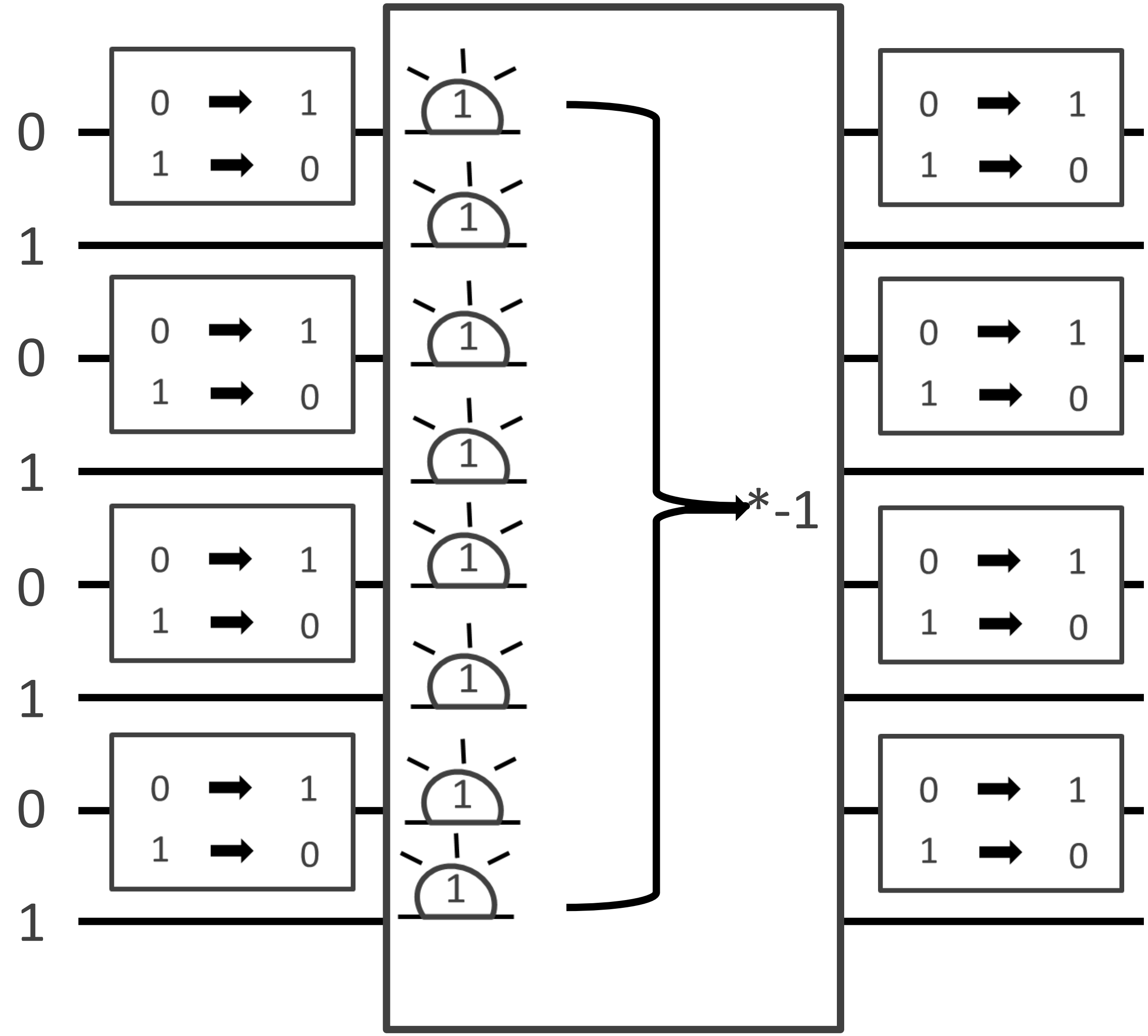

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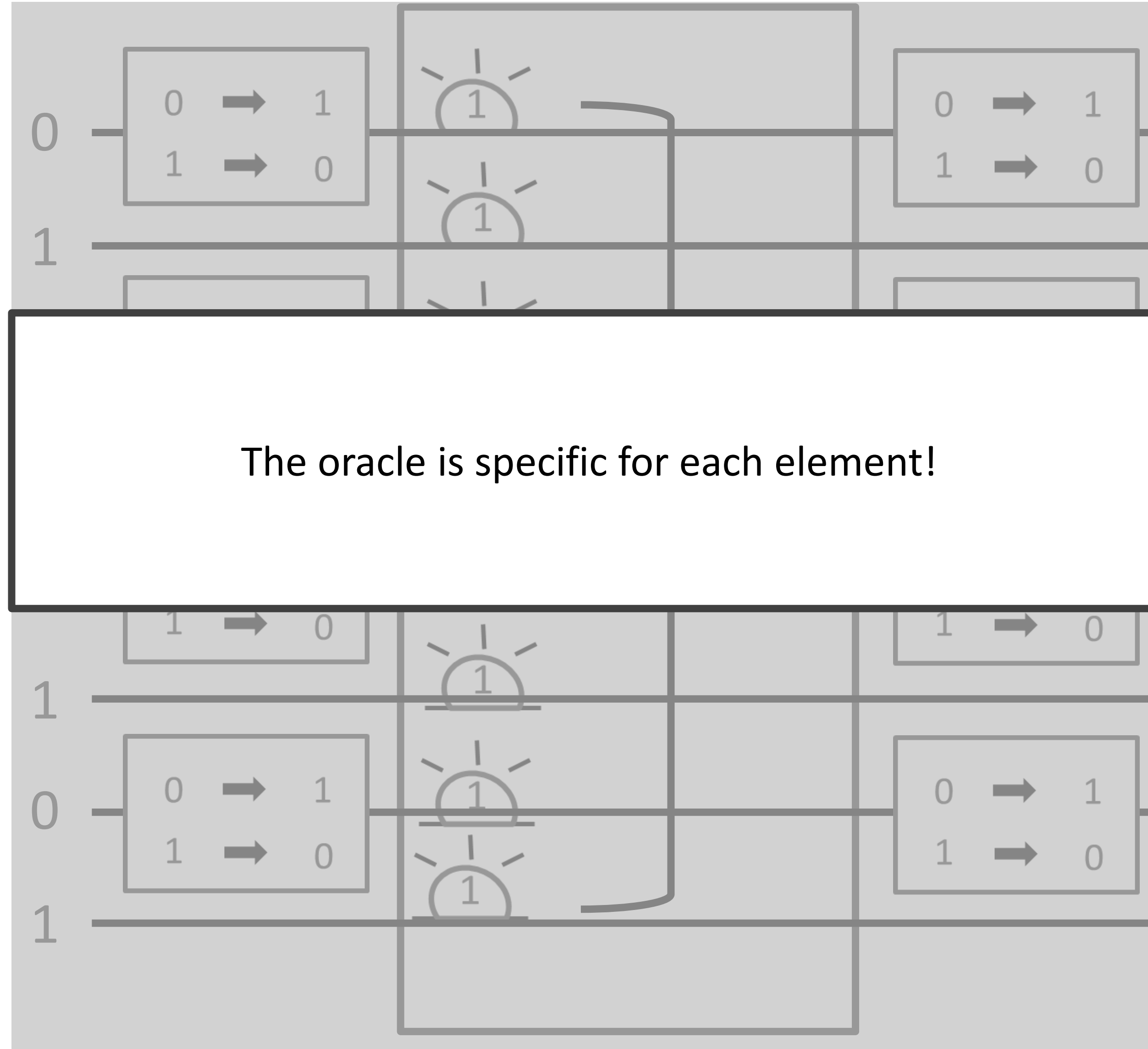
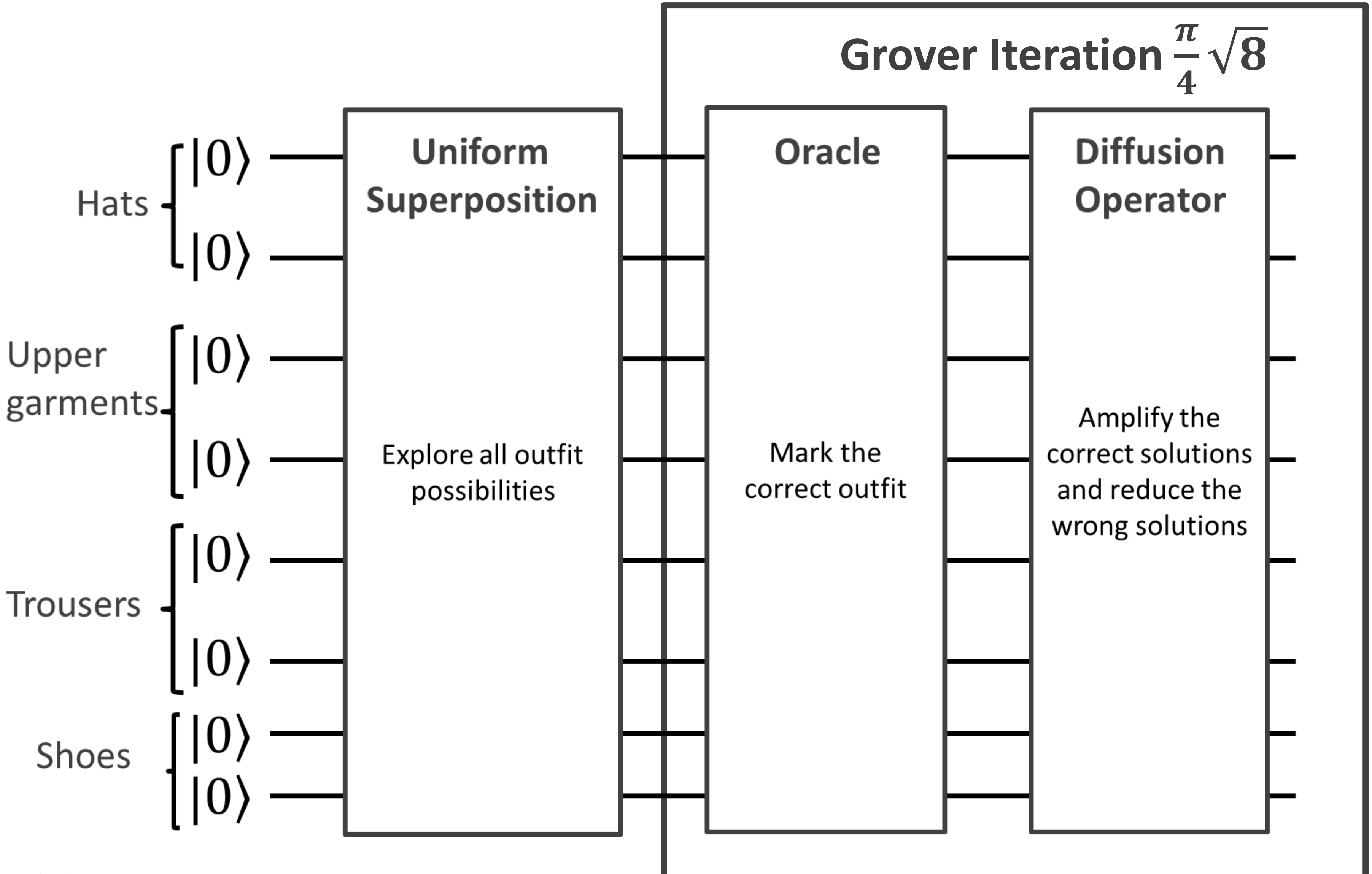
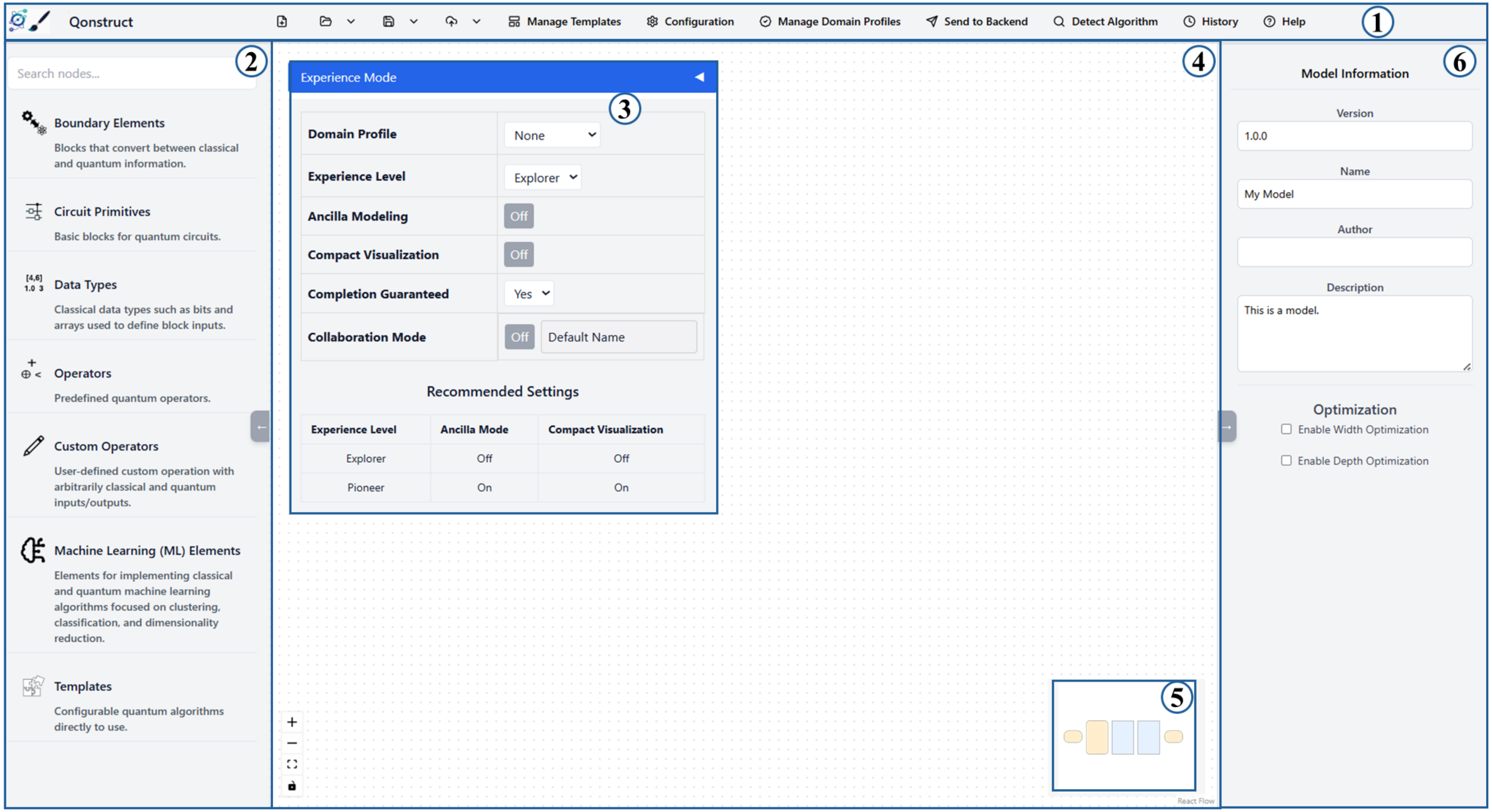


Task for Today - Grover‘s Search Algorithm

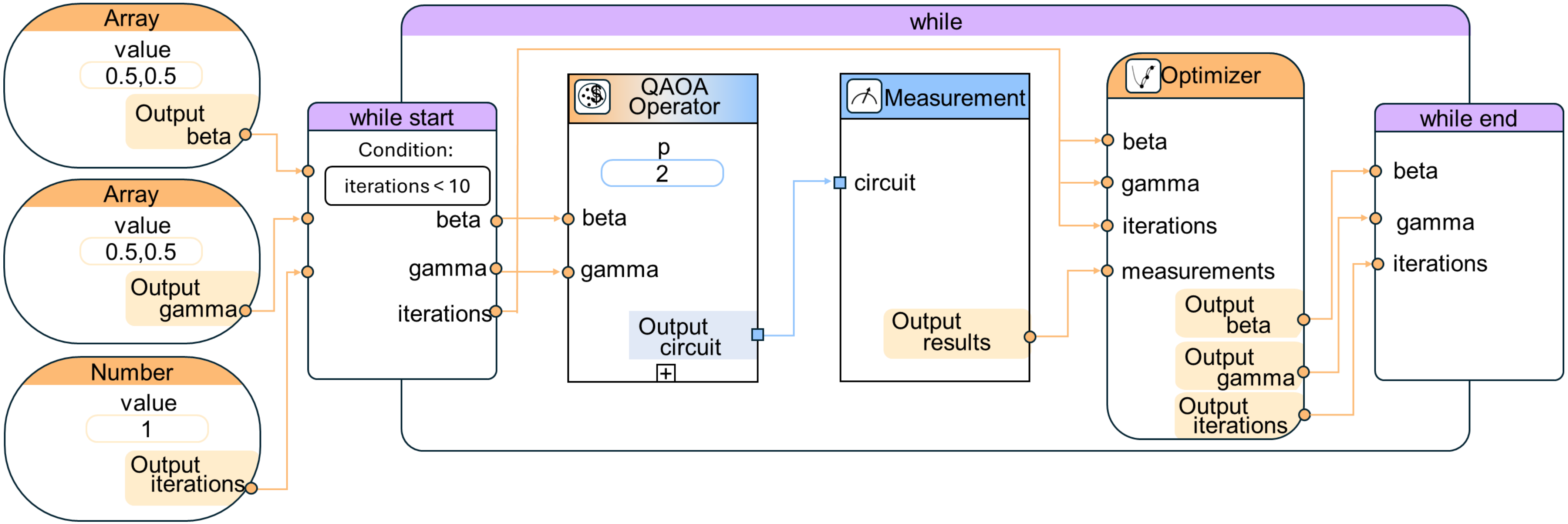


Qonstruct Overview



Legend:
① Toolbar ② Palette ③ Experience Mode Panel ④ Canvas ⑤ Mini Map ⑥ Properties Provider

Modeling and Execution of Quantum Algorithms



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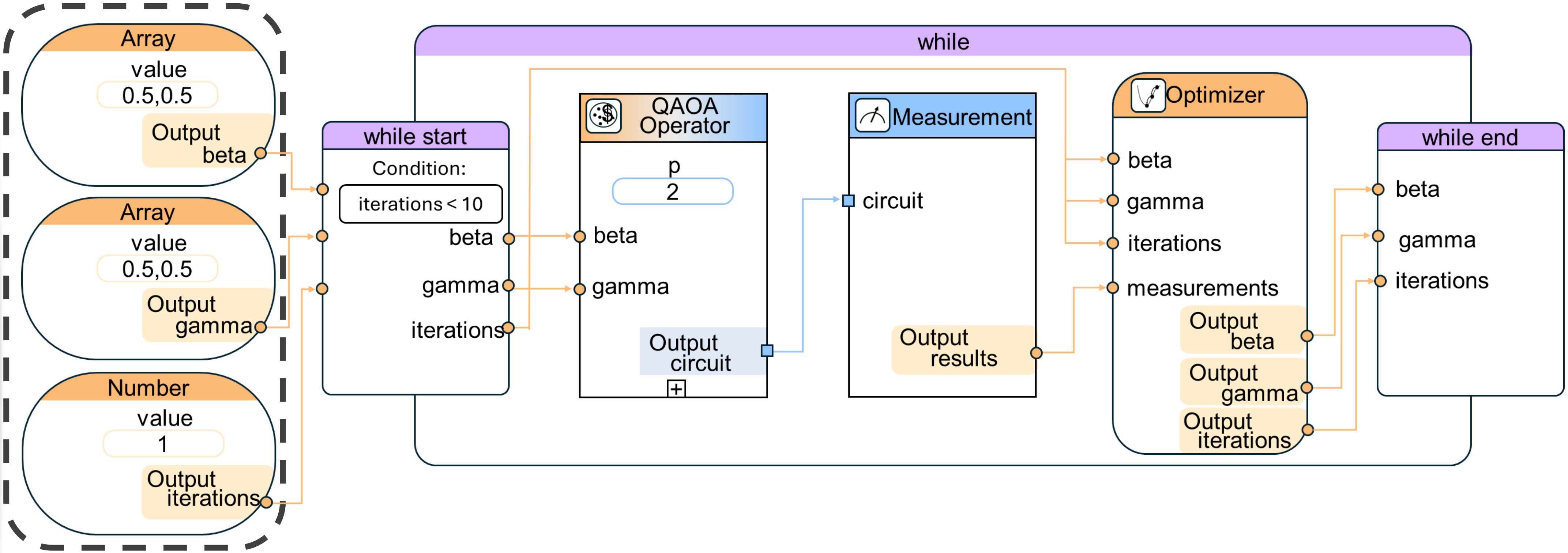

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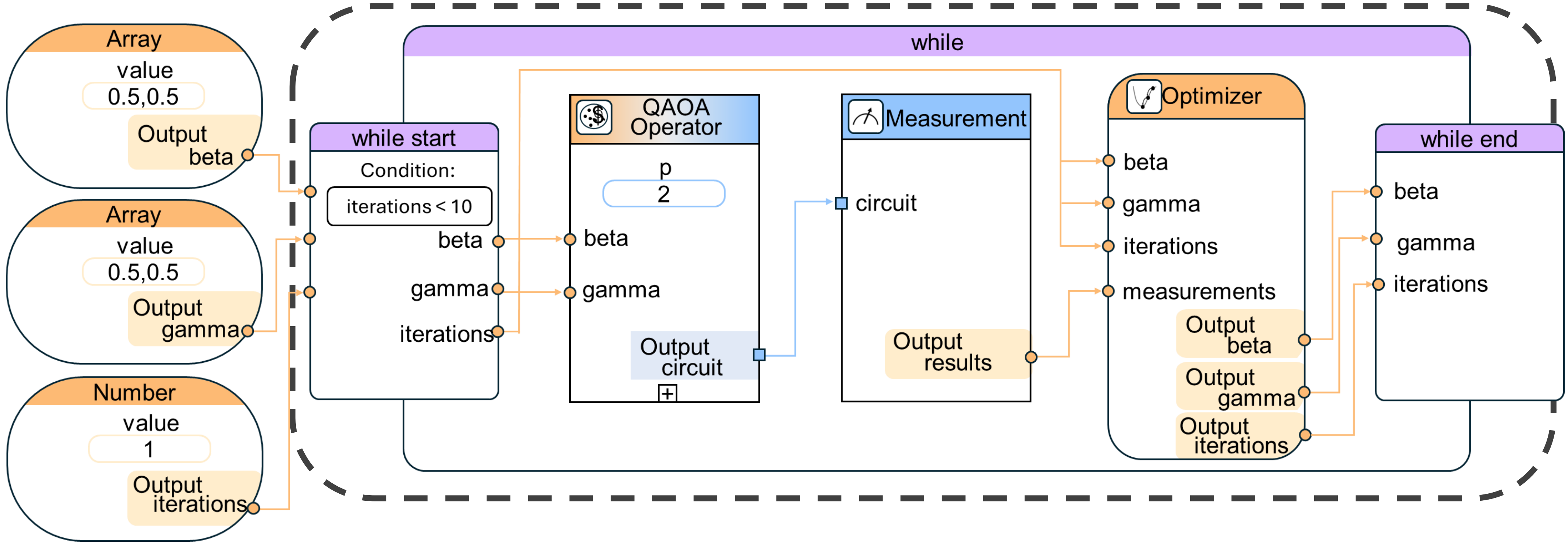

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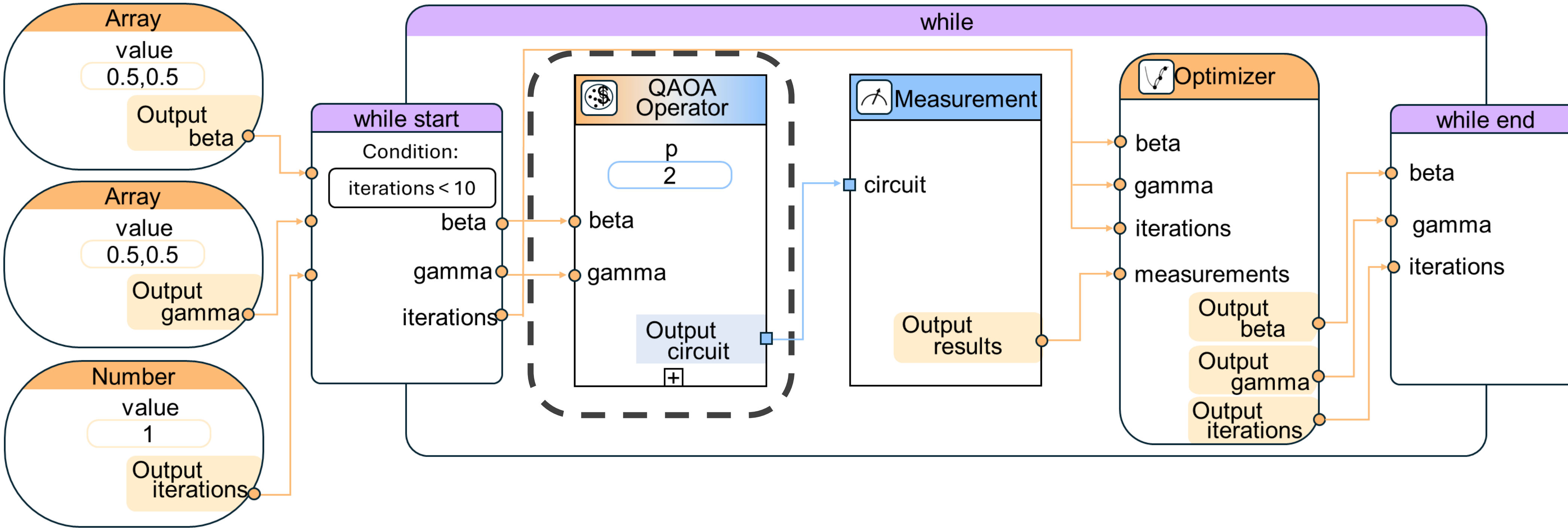

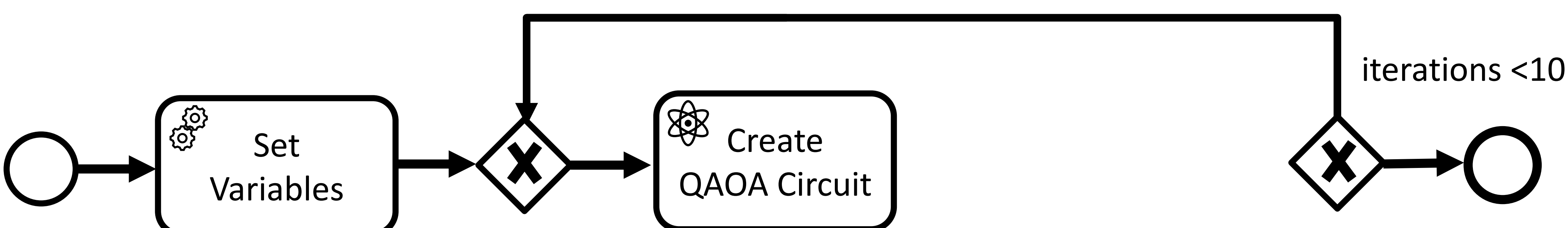

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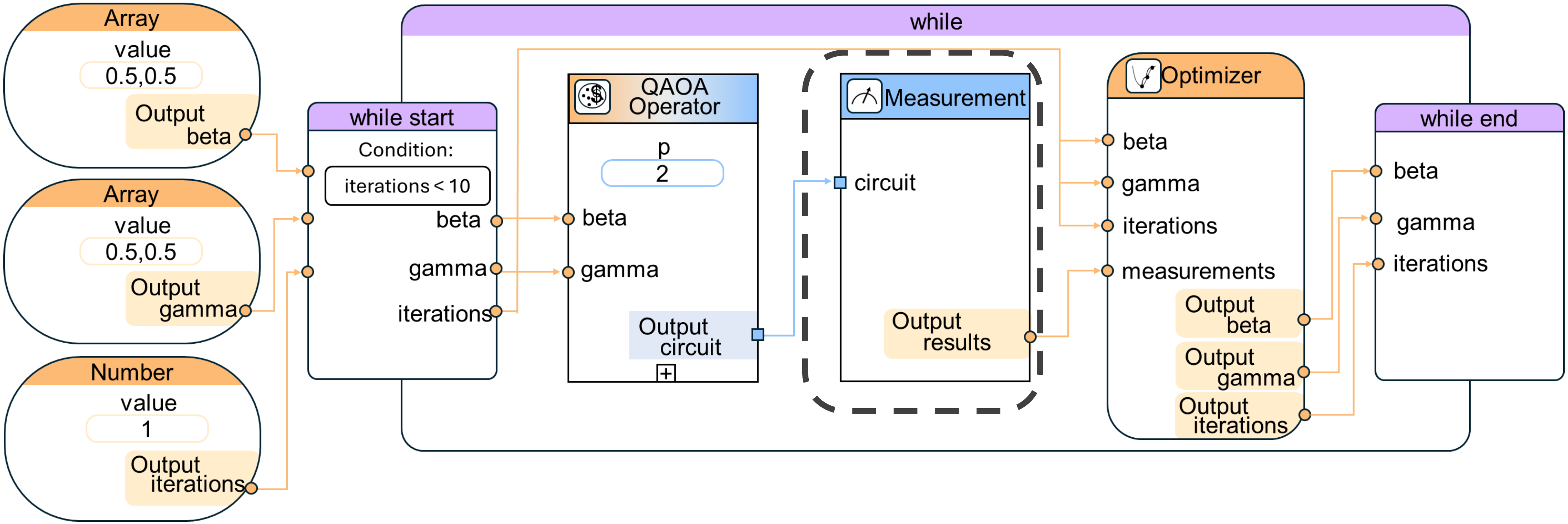
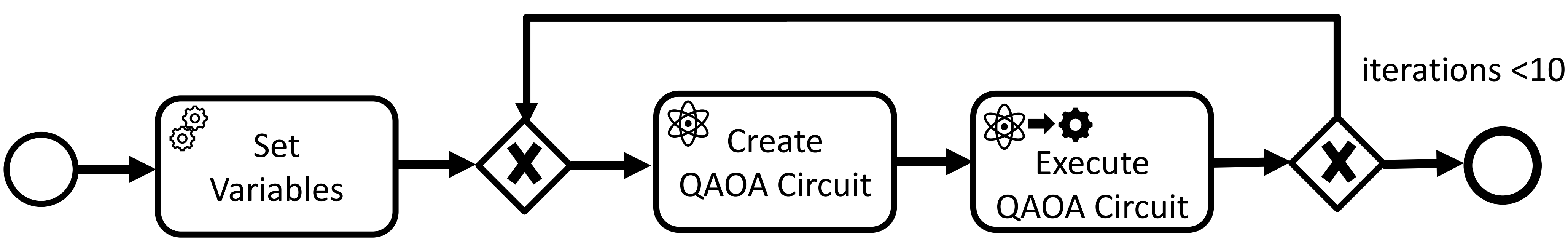
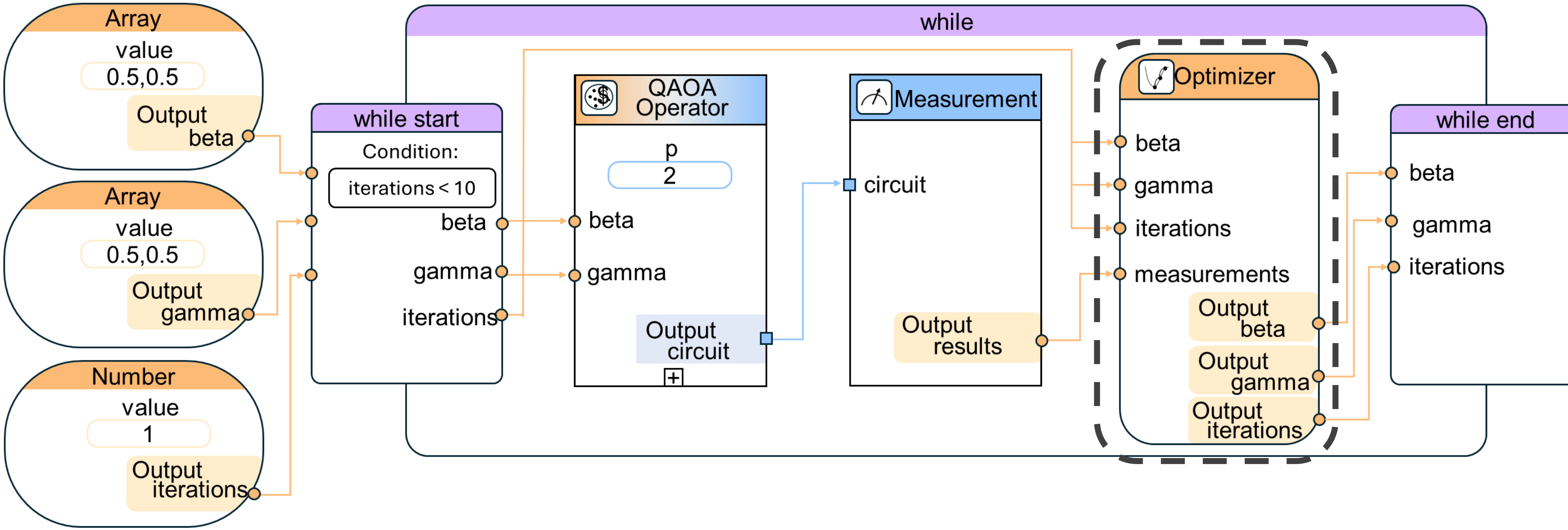
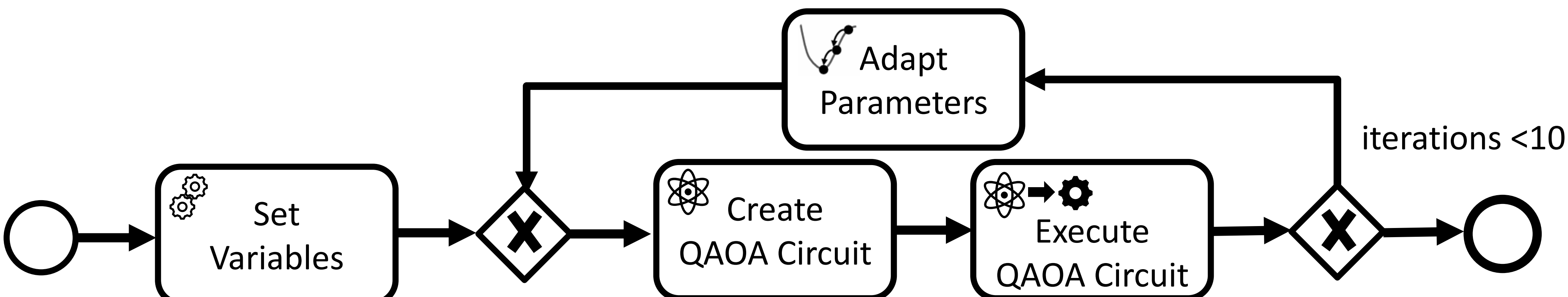




Modeling and Execution of Quantum Algorithms





Thank you for your attention!